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Contact No. 1805

Intellectual Property Disclosure Form

1. **Title of the invention:** A system to facilitate agreement when multiple individuals have conflicting interests and limited disclosure

2. **Innovator(s) who have contributed or conceived an essential element of the invention, either independently or jointly with others during the evolution of the technology concept or reduction to practice:**

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Inventors are requested to provide their Full Name (without initials**), Position (**e.g Professor / Associate Professor / Assistant Professor, Post Doc / PhD / M. Tech / B. Tech Student**), Phone Number (**Personal**), Email ID (**Official along with Personal**)*

Students are advised to provide their **home address as well.*

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3. Non-Confidential description of the invention in layman's Language:

A. Abstract in 100 words

In a low-resource setting, the same device and application get used for all kinds of tasks. These include personal, private, public, professional, confidential, etc. As reliance on agentic systems and overlapping care/social/professional circles increases, there is an increasing risk of crosstalk due to human and/or machine error. Existing systems rely on identity verification, access control, and secure communication channels, but do not provide enforceable guarantees on how data is used once shared. The disclosure introduces a system comprising nodes with secure enclaves that operate across multiple modes. The system enables AI-driven companions to perform information discovery, selective disclosure, and collaborative document drafting in environments where participants have conflicting objectives and limited willingness to share information.

B. Use Case: A system for secure multi-party interaction and agreement under limited disclosure

In real-world scenarios, individuals and organizations interact using shared devices and applications across multiple contexts such as personal, professional, and confidential environments. In such settings, a user (U_1) may receive sensitive information from another user (U_2) while simultaneously interacting with a third user (U_3) or another organization. Existing systems rely on application-level separation or access control, which cannot guarantee that data received from U_2 will not be accessed, inferred, or leaked during interactions with U_3, especially when AI agents assist in communication, drafting, or decision-making.

The disclosed system introduces a node (N_1) operating in dual modes, wherein a secure mode enforces hardware-level isolation of data, execution, and communication restricted to a specific interaction context ("circle"), while a utility mode enables flexible multi-context interactions. When U_1 receives data from U_2 under secure mode, the system ensures that the data remains confined within that mode and is not accessible to processes, agents, or interactions in utility mode or other circles involving U_3.

In multi-organizational settings, where participants have conflicting objectives and limited willingness to disclose information, the system enables AI-driven companions to assist in document drafting and negotiation. Each companion selectively discloses information based on quantified disclosure advantage, inferred objectives, and consent constraints, while operating within hardware-enforced boundaries that prevent unauthorized data propagation.

This enables secure collaboration, negotiation, and agreement formation across individuals and organizations without requiring full trust or complete transparency, while ensuring that sensitive data remains protected even in the presence of autonomous agents and overlapping interaction contexts.

Please consult the examples provided in Annexure-1 for filling this section

C. Keywords

Information and Communications Technology, Social Artificial Intelligence, Healthcare, Emerging Technologies, Multi-agent system, Data reconciliation, Patient adherence, Care circle, Conflict resolution, Insurance processing.

Please note that the above keywords will be utilised by the IPR Cell for preparing the Patent Search Report

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Note: *Please note that the above Information alone will be circulated to several agencies for technology commercialisation purposes once the patent is filed. Thus, the fields should be self-explanatory to highlight commercialisation potential.*

4. How does this invention relate to new processes, machines, compositions of matter, etc.? Please cover the following points:

(a) Describe the invention in detail for technical evaluation. Please use additional sheets for sketches, drawing, photographs and other materials that help to illustrate the description.

The invention is meant to mediate interaction among members of 'circles' (क) (including care circles, professional circles, and social circles) among individuals and outside, inside and between organizations. It supports and mediates private interactions among the members of a circle by assigning each of them personalized companions. Each companion (C) comprises a user node and a cluster of AI agents that collaborate to optimize the performance of the companion. The communication between the companions is mediated by a central orchestration engine (Ω) that comprises the orchestration agents. The orchestration agents run on the server node(s). The agents in a companion are logically coupled with the user's node and run on either the user node (N_u) or server node. The system, therefore, requires a minimum of 3+1 nodes: target's node (patient's node), the support's node (caregiver's node), the expert's node (clinician's node), and the orchestration server (N_Ω). A user interacts with his or her companion through the node dedicated to him or her. The user nodes have partitioned memory, storage, and compute.

In a real-world scenario, circles overlap, organizational roles and limits are enforced, and roles emerge. This leads to the emergence of organizational subnetworks, or 'orgnets' (O) that interact with each other. An orgnet is governed by the orgnet coordination engine (Ω_c) and server ($N_{\Omega c}$). The coordination engine decentralizes the system to increase the control and privacy of an organization on both the data and agents. It acts as a gateway between the orchestration server and the companion node. The goal of a coordination engine and companions in an orgnet is to protect the interests and maximize the gains of the members of the orgnet.

The system utilizes a distributed network topology comprising the following distinct hardware elements:

1. User nodes (N_u): These are physically distinct computing units that comprise
    a. A bidirectional visual interface: a display module integrated with a touch-sensitive layer and haptics,
    b. Audio & text input modules: A microphone and digital keypad configured to capture the raw natural language, image, and audio-video data streams from a user's node ($ड_U$). The data streams include the user's identity (I_U). Similarly, data about the subject ($ड_s$) and the subject's identity (I_S), with data about the object ($ड_o$) and identity of the object (I_o) are also included. The subject and object can be mapped to other users using the system. The data streams are also tagged with the timestamp and geostamp whenever an input is received from a user.
    c. Wireless communications modules like cellular, Wi-Fi, Bluetooth, LiFi
    d. Wired communication module accessible through serial bus
    e. The personal subnet manager that functions as a switch or a hub to identify the connected telemetry devices (like smartwatches, smart glasses, digital signatures, digital identity cards, mobile phones, digital thermometers, digital glucometers, electrocardiography machines, etc.) and tag the data streams with the patient's

identifier. The personal subnet manager manages multiple devices being connected at the same time or at different points in time. In the latter case, it will have to retain the identity, drivers, and data formats used by each telemetry device.

f. Camera module with photo and video support
g. Ports for connecting to other devices.
h. Memory and storage modules

2. Central server (N_Ω) and coordination servers: High-performance computing units that comprise the following:
    a. Partitioned memory and storage: Configured to maintain isolated state vectors for each user, companion, conversation, document, and circle to prevent cross-contamination of subjective data.
    b. Processor: One or more processing units configured to execute the instruction transcoding and forking protocol, capable of taking a single command from the one node and outputting one or more distinct execution protocols for the other nodes.
    c. Network module: A modem and necessary ports required for connecting to the companion nodes over the internet using protocols and frameworks like IPv4, HTTP, FTP, WebRTC, etc.
3. Common observation nodes (N_g): These include microphones and web cameras in conference rooms, CCTV cameras, and other devices that collect data streams of multiple users and share them with the server. The server maps them with the relevant users and circles actively or on demand. The input modalities include text, audio, images, videos, documents, messages, alerts, files, etc.

A user node is used for multimedia functionalities, including but not limited to calling and accessing the internet with or without restrictions. The user nodes are also used to intercept direct user-to-user communications like phone calls, messages, and internet messages. These intercepted streams are used as inputs to the personal agentic cluster. After analysis, the system notifies the authorized circles and their members about the relevant events and actions, if required. The autonomy of the user's companion is decided by configuring it directly, by granting control permission to another user's companion, or by having a vote in a circle. Additionally, the type of circle and the type of information and action further affect the protocols that the companion will follow.

A node N_1 proves its trustability to another node N_2 by demonstrating controlled operation, verifiable integrity, and auditable data handling.

A node (N_1) operates in multiple modes of operation, including a utility mode and a secure mode. The utility mode allows the user's companion to practice autonomy and select the active circle as required. The secure mode restricts N_1's operation to a single circle at a time. It enforces isolation at application, operating system, and communication levels, including restrictions such as blocking screenshots and preventing data flow between secure mode and utility mode. N_2 retains control over data shared with N_1, including the ability to restrict, modify, delete, or track its propagation through the relevant companions.

Utility and secure modes operate in parallel. The secure mode does not disable the utility mode but ensures that processes, memory, and data within secure mode cannot be accessed, modified, or influenced by processes running in utility mode. Any data received or generated within secure mode remains confined to it and cannot be transferred to utility mode. If a handshake is made in the secure mode, communication only proceeds in the secure mode. Companions dynamically select and operate within the relevant circles (क) based on task, and individuals involved. The same companion

participates across multiple circles; however, its access to data and permissible actions are constrained by the mode (utility vs secure) and context (the topics and circles) in which it operates. While a single companion instance may operate across both utility and secure modes, memory and data access are strictly segregated. Information available in secure mode is not accessible to the companion when operating in utility mode, and no implicit memory transfer occurs across modes. Information may be shared across multiple circles within utility mode subject to consent and role-based permissions. In contrast, secure mode operates with a restricted and fixed information set, and does not inherit or share contextual data across circles unless explicitly permitted and transferred under controlled conditions.

The secure mode also has two sub-modes of operation. The first is true isolation. In this mode, no data from the utility enclave can move into the secure enclave. This protects the user's private information from contaminating a sterile enclave. The other mode allows the user and companion to identify information from the utility enclaves that, upon disclosure to a circle coupled with a secure enclave, provides a significant advantage. The same information is marked as confidential and can't be shared with another companion (C_2) without explicit user consent. Upon reaching the second node (N_2), the second companion can not disclose it to its user or move it to its utility enclave.

A security module enforces integrity across hardware and software layers. It maintains a set of physical and system state variables that reflect device integrity, firmware state, and execution environment. Any physical tampering, such as opening a screw or replacing a component, or any deviation in system state, updates one or more of these variables in a non-reversible manner. These state variables are stored in secure storage and are used to compute a security measurement by combining them with system state information. This measurement is bound to a hardware root of trust and is included in the signed attestation done by the orchestration and coordination servers (Ω, Ω_c).

During communication, N_1 presents this attested measurement to N_2 as part of a handshake. N_2 verifies the attestation against expected states across multiple layers, including application, operating system, firmware, hardware, and network conditions, to determine whether N_1 is in a trusted state. N_1 shares its validated level of security with N_2 across these layers using certificates and signatures through the orchestration server (Ω). This enables verification of trustability. All data access, modification, and sharing events are tracked using blockchain-based mutation logs to ensure transparency and auditability.

A node N_1 can prove its trustability to another node N_2. To prove the trustability, the below are required.

1. A node (N_1) operates in multiple modes. The modes include utility mode and secure mode.
    a. The utility mode allows the user's companion to practice autonomy and select the active circle as and when required.
    b. The secure mode enforces isolation of the N_1's operation to a single circle in addition to application and OS-level restrictions (like blocking screenshots). N_2 can also control data shared by N_2 to N_1, get it deleted or modified, or track where it is shared.
    c. During parallel operation of utility and secure mode, the secure mode blocks the communication, inputs, and outputs of the utility mode. The secure mode is only meant to ensure that the processes in the utility mode can't interfere with or tamper with the processes and data of the secure mode. Any operation in secure mode is guaranteed to stay in the secure mode.

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2. A security module—In addition to software-level security, it ensures hardware-level security by storing physical state as a variable. For example, if a screw is ever opened or a chip is replaced, the state gets updated permanently and is stored in secure storage. This change in state leads to a change in the security hash code, which is computed by using a time-varying signal from the orchestration/coordination server. This state is then used during handshakes during trust assessment.
3. Security enclaves—Partitioned memory and storage and optionally partitioned compute. To provide isolation of secure information and processes.
4. A node (N_1) sharing its validated level of security with another node. These security levels include application level, network level, OS level, firmware level, hardware level, etc. The confirmation of validation is done by sharing through orchestration server-synced certificates and signatures. This is done to prove the trustability of a node.
5. Blockchain-based data mutation tracking.

Two devices in the system/network can do handshakes, authorization, and authentication through multiple means, including manual input, scanning QR codes, the wired or wireless method, etc. A companion node can be added to a circle through such a handshake and thereby connect with other members of the circle.

One handshake method relies on common observation nodes (N_g). The companion (C) queries the orchestrator (Ω) by providing the identity of the user (U), location, and timestamp. The orchestration engine (Ω) contacts the relevant coordination engine (Ω_c) and the relevant common observation node (N_g). Ω, Ω_c and N_g share information with each other to match the biometric footprint of the user and thereby complete authentication. This can be used to connect user U_1 or companion C_1 from O_1 to O_2, U_2, or another device on the network N_k.

Another handshake involves 2 users. U_1 and U_2 and a common node N_g. If C_{U1} has completed a handshake and for some reason C_{U2} fails to complete the handshake with either C_{U1} or N_g, N_g will take implicit or explicit permission through C_{U1} to query the identity of U_1 and then tag the data stream for U_2.

A third handshake method requires device N_1 to emit its identity and a time-varying, server-verifiable signature using a physically isolated trust channel. A user node N_{C2} is then used to capture the received signal using specialized or general-purpose optical sensors. N_{C2} then verifies the identity of N_1 through the orchestration and coordination servers (Ω and Ω_c). Beyond this, the handshake can securely proceed using time-varying private key encryption using any of the currently used protocols. Such a handshake requires trustworthy processes and hardware.

Depending on the privacy settings of the user U_1's companion (C), the trustworthiness of the coordination engine (Ω_c) and the trustworthiness of the common observation node (N_g), the identity of the user (I_U) or a function to confirm the biometric pattern of user U is shared with N_g or Ω_c by Ω for authentication. In case of low trust on Ω_c and N_g, Ω is used for analyses and authentication.

A special scenario is one in which one user's U_1 interacts with an individual U_2 where U_2 is not on the system. In such a case, U_1's companion C_{U1} or a common observer node N_g captures the interaction between U_1 and U_2. In such a scenario, a dummy user U' is added to a new or existing circle (क). The information captured by C_{U1} or N_g, therefore, refers to U_2 as either a subject or an object. In a future interaction, C_{U1} may look up U' in its memory.

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A user can copy/move information from one device to another through multiple ways.

1. Looking or pointing - Look at a target piece of information. Click and hold a button on a device that is connected to the headgear. The click will lead to the camera facing the information and the camera facing the eyes to identify what is being looked at. Alternatively, the user can use a pointing device like a stylus for the same. The user will then have to look/point/touch at the device that requires the information and let go of the button. The camera in the heads-up display will identify the receiving device, share the information over the network, and also share the location where it is to be placed/inserted.
2. Selecting initial information by interacting with the first device. Then using a control device to select the second device and share the information with it.
3. Interacting with a piece of information on one device using a second device, like a stylus, and then interacting with the third device using the second device.

Each user node in the system allows for both sending and receiving information. Information includes text messages, voice messages, video recordings, real-time audio/video calls, files, data streams from telemetry data, etc. They may also have the ability to control more nodes in the network, for example, giving commands to printers.

These nodes (including care-o-meters and the clinician nodes) are used by the users (patient, caregiver, and clinician) to interact with the personal agentic cluster of their companion. These agents perform tasks including but not limited to the below. There may be unified, dedicated, or multiple agents for each of these tasks. These agents run on user devices and/or on servers.

1. Identifying the best response to the user's input
2. Executing Context-Aware Query Protocols to minimize State Vector sparsity. This is done by asking probing questions from the patient and the caregiver to uncover new information and resolve conflicts identified by the orchestrating agent.
3. Parsing and storing information while mapping the source, disclosure date-time, validity date-time, subject, object, revisions, format, originality, etc.
4. Deciding what information needs to be sent to their corresponding users
5. Blocking and/or defering disclosure of information to a user depending on the content of the information and the readiness of the user to accept the same information.
6. Deciding if another user's companion needs to be contacted
7. Deciding what information needs to be sent to or sought from another user's companion
8. Checking if user consent is available for forwarding data
9. Seeking explicit consent through the terminals when needed
10. Implying consent from the user's configuration, conversation tone, topic of discussion
11. Assessing the sensitivity of the topic and choosing to override sharing/escalation requests by the escalating agents. For example, strategy, lay offs, financial topics, mental health, topics related to sexual health and life, crime, non-disclosure agreements, etc.
12. Assessing the criticality of situations (including but not limited to risk of self-harm and possibility of an accident or medical emergency) and choosing to escalate even if the topics are sensitive.
13. User's past history collection
14. Conversation topic management: choosing the next topic
15. Creating a representation of the user and the circle
16. Creating a representation of the users' physical environment
17. Identifying the names of people mentioned in user messages and mapping them to their user and care circle.
18. Understanding and generating diagrams, images, videos, volumetric data, audio, etc.

19. Strategizing, planning, creating, and scheduling longitudinal interventions depending on the category they fall in: decision, process, behavior, and negotiation. Creating tasks, milestones, timelines, reminders, etc.
20. Strategizing in favour of the user's personal goals
21. Schedule tasks and messages to be sent in the future
22. Communicate with APIs of different devices and services outside this system, for example medical devices, mobile phone apps, smartwatches, smart water bottles, medication boxes, etc.
23. Coordinate communication and actions among the above agents, the orchestrating agents, and/or the user.
24. Resolving conflicts between agents within a companion (i.e. cluster of personal agents). This is be done through various approaches like ranking, reasoning, heuristics, etc.
25. Keeping track of social factors like relationships, closeness, openness, etc.
26. Anonymized escalations
27. Deploying structured interaction modules to standardize user input patterns: Asking questionnaires and providing psychiatric therapies
28. Tracking which target (e.g. patient), and circle, the user (e.g. caregiver or clinician) is currently interested in.
29. Identifying the role a user is playing by writing a message (target, patient, support, caregiver, expert, clinician, etc.)
30. Flagging critical but sensitive/private information of one user as confidential so that it can only be shared with the orchestrating engine and a second user's companion. The confidential information can never be shared by the second companion with its user. This is only for strategizing.
31. Delegate tasks like asking probing questions, questionnaires, and instructions to other users' companions, and then collaborate with them to create a report.
32. Sharing, analyzing, editing, and comparing different types of files. Collaborating and commenting on drafts. Filling, sharing, signing, and approving forms, records, and files.
33. Analyzing the content, comments, and changes to records, files, forms, documents, and summaries and then providing feedback and suggesting changes.
34. Manage formal and informal channels of communication.
35. Interface with organizational systems like the enterprise resource platform and the customer relationship management system.
36. Discovery of information from various sources like private and public databases, including the internet, neural networks, APIs, or direct inputs from users like messages, files, videos, telemetry devices, observation devices, etc.
37. Analyzing the recordings and live streams of audio/video of in-person, telephonic, message-based, and online interactions among the users and other individuals as captured by the nodes. By configuring the agents to recognize the voices of circle members, the transcriptions can be mapped to the speakers, listeners, and subjects. The interactions can be summarized, and actionable items can be decided using the agents.
38. Keep track of the location and schedules of users and share them with other users within and outside the organization/circle as and when required.
39. Analyze audio/video feed from common nodes (like CCTV and other surveillance devices) and share it with the relevant circle. Switch circles that receive the video feed and insights by analyzing who is in the video feed. Track patients as they move between fields of view of different cameras. Assess behaviors and other actions based on the video feeds.

The agents in the orchestration engine (and coordination engines) coordinate the different companions and facilitate communication between them. Their primary goal is to guide the companions towards the collective goal of a care circle. The agents in the orchestrating engine manage activities like the below. The orchestration engine may include personal agents.

1. Storing user identities
2. Tracking professional and circle configurations
3. Tracking consent configurations—availability, level, revocations, chains, etc.
4. Keeping track of relationships
5. Keeping track of roles within circles
6. Keeping track of contributors and contributions to a document, file, or record.
7. Keeping track of the changes and versions of a record, file or document.
8. Selecting message receivers or choosing to broadcast information across a care circle.
9. Personalizing message content according to the receiver.
10. Scheduling and sending notifications to users through companions.
11. Receiving message requests and instructions from one companion and forwarding them to another companion, with or without changes.
12. Inverting user roles—a patient becoming a caregiver and a caregiver becoming a patient. Similarly, a boss becoming a subordinate and vice versa.
13. Identify and understand relationships, family dynamics, and workplace dynamics. Adapt by setting the tone and strategy of communication, documentation, negotiation, and instructions.
14. Decide the strategy for the companions and users in a professional or care circle.
15. Track progress with the goals of the professional or care circle.
16. Edit statements requested by one companion or its user so that they suit the perspective of another companion's user.
17. Block information from one companion to another if it can damage the relationship of the sender and receiver.
18. Store the secrets of one user and instruct another companion to strategize the conversation with a second user such that the first user can benefit without a privacy breach.
19. Connecting to various third-party APIs and tools.

Training through simulations and reinforcement learning through human feedback

As the system is novel, it is not possible to train a powerful neural network on a prior dataset because no such dataset exists. Therefore, the only way to train the model before real human usage is by simulating human users that perform different scenarios and roles. These scenarios can also be simulated by generating the user responses at runtime. For a more realistic simulation, the users are simulated by generating dialog graphs through zero-shot and few-shot learning. The simulated user messages can be even more realistic if handcrafted dialog graphs are leveraged. The dialog graphs also provide test cases through which each agent, action, and model can be validated objectively on predefined interactions and parameters. In addition to simulations and dialog graphs, training for negotiation, adversarial scenarios and strategy can also be done by making the system play social strategy games that involve negotiations, for example, So Long Sucker.

Evaluation can be done on different scenarios like care circle dynamics, power plays, triangulation, wrong clinical advice, non-adherence, addictions, breaking the law, dishonesty, hiding information, convincing, disease progression monitoring, follow-up, adherence support, caregiver burnout, medical emergencies, thwarting tactics, sales, negotiation, patient education, seeking answers, administrative tasks, document drafting, approvals and objections, conflicts of interest, fights, mistakes, losses, payment delays, unavailability of funds, information retrieval from different internal and external databases, care circles and professional circles overlapping with each other, etc.

Evaluation of real and simulated interactions is done by human experts. For large-scale evaluation, judging agents judge the actions and messages of the agents in the system.

The judging agents and human evaluators judge the simulated users for diversity and realism. The evaluations include both quantitative and qualitative aspects.

For automated testing and training, the qualitative aspect, i.e., comments and feedback on the conversation/action, are used to update the prompt and then perform a split test using the two versions. Similarly, quantitative aspects are used for fine-tuning existing models.

An orgnet (O) can use agents trained on private data owned by the orgnet. The coordination engine oversees the creation of these private agents and their fine-tuning. An orgnet also has hierarchies and escalation matrices such that a user and companion can receive the relevant information at the right time, after due diligence, and act accordingly. Furthermore, the exit of one user will lead to the handover and/or deletion of information possessed by the user's companion. This is enforced by the coordination engine by destroying companions along with associated data.

A coordination engine also has the ability to create a standalone companion (C_s). The corresponding physical node (N_{CS}) is a portable device that is disconnected from the system once created. This N_{CS} includes a compute unit, memory, storage, and connector (USB) but no networking or recording capability. An N_{CS} is configured upon creation to store the user's identity, signature, pre-approved organizational and personal data (the objective, instructions, details about the circle and its members, biometrics, etc.), and the relevant agents. An N_{CS} is meant to be carried into a secure boardroom that only has non-network and non-storage devices like input/out peripherals. These devices and N_{CS} are connected to a Compute hub (N_h) that runs the companions. This allows the boardroom discussions to be supported by agentic AI companions in a confidential manner. The N_{CS} can be configured to delete all of its contents after a set time and the outputs of the meeting - minutes, reports, contracts, etc can be stored on the N_h to be moved to another device. In this process, the N_{CS} and N_h work together to identify faces and voices captured by the peripherals, tag them to the correct user and process the transcriptions using the relevant companions.

Algorithm for multi-source information discovery and corroboration

Step 1: Circle and Agent Network Instantiation - A user (example patient) initiates a secure, goal-oriented group (a "circle") by inviting other members, such as a caregiver or doctor. Upon joining a circle, each member is assigned a dedicated cluster of personal AI agents (Companion). The patient defines the initial roles of care circle members and granular consent levels for information sharing. These are managed by a consent management engine, which handles the granting, modification, and revocation of permissions within the circle. The agents in the consent management engine are divided among companions and the orchestration/coordination engine. This network operates over one or more communicatively coupled processing servers, which can be centralized or distributed.

Step 2: Multi-Source Data Ingestion and State Modeling - Each member of the circle conversationally interacts with their companion, reporting their own observations (e.g., symptoms, adherence) and their observations of other members. The agents independently process these multi-source data streams, continuously updating a comprehensive, high-dimensional state vector (S_t). This vector serves as the complete context for all AI decision-making, capturing the circle's clinical, behavioral, social, and conversational states at any given moment.

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Step 3: Multi-Agent Discrepancy Reconciliation - The system's core Multi-Agent Reconciliation Engine analyzes the state vectors (S_t) to compare data streams from different users (e.g., the patient's report R_p vs. the caregiver's report R_c). When a factual conflict is detected (e.g., conflicting reports on medication adherence), the engine flags a discrepancy. To address this discrepency, the system generates a neutral, mediated clarifying questions directed at one or more members. This questions are designed to elicit more detailed information to establish a reliable ground truth. Depending on the level of consent and sensitivity of the topic, the final report is generated transparently or without revealing the source of the conflicting information, thereby preserving trust within the circle.

Step 4: Actionable Insight Delivery - Following reconciliation, the system generates a unified and detailed assessment of the user U_1's behaviour and overall state. The tangible technical output is the rendering of this reconciled state on another user U_2's interface, providing a consolidated, at-a-glance view of patient status, active conflicts, and critical alerts. This enables the human expert (clinician) to make timely and effective interventions. For events assessed as having high criticality, the system is configured to trigger an immediate escalation to the appropriate professional, bypassing standard workflows to ensure safety and prompt care.

Algorithm for Cross-Organizational Negotiation via Companion Agents

1. Definitions

   Let:
   O_i = Organization
   O_init = Initiating organization
   O_target = Counterparty organization
   $U_{i,j}$ = User j in organization i
   $C_{i,j}$ = Companion agent of user $U_{i,j}$
   R_i = Set of roles within organization i
   $r \in R_i$ = A role
   D_k = Draft document version k
   $\Pi_{i,j}$ = Privacy policy of user $U_{i,j}$
   Γ = Global negotiation state
   Ω = Orchestrator
   Ω_i = Intra-organizational coordination engine for O_i
   Ar = Arbitrator (optional trusted entity)
   D_internal = Organization-specific internal data
   D_external = Public or externally available data
   D_collected = Data collected from users
   O_goal_i = Organizational objective function for O_i
   $U_goal_{i,j}$ = Personal objective function for $U_{i,j}$
   G_common = Shared objective function across participating organizations
   G_declared = explicitly stated objective of user U
   G_inferred = inferred objective of user U or U's companion C_U based on observed behavior, historical interactions, and external signals. The inference may be done by the C_U or another user's companion $C_{U'}$.
   k = structured clause derived from one or more data elements d
   K = Set of all clauses or statements in the draft D_k
   d = data element

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C_i□ optimizes U_goal_i□
Ω_i optimizes O_goal_i. It enforces intra-organizational consensus.
Ω optimizes G_common subject to constraints. It drives inter-organizational consensus.

2. Core Primitives

a) Consent Function
CONSENT(U_i□, data, target, context) → {ALLOW, DENY, CONDITIONAL}

b) Implicit Consent Derivation
IMPLICIT_CONSENT(Ruleset R, context Γ) → inferred permissions

c) Privacy Filter
FILTER(data, Π_i□, target_scope) → redacted_data

d) Conflict Detection
CONFLICT(x_1, x_2, …, x□) → {TRUE, FALSE, ambiguity_score}

e) Mediated Update
UPDATE(D□, inputs, constraints) → D□$_{+1}$

f) Utility Function
UTILITY(x, actor) = SUM over all data pairs (d_1, d_2) of INTERACTION(d_1, d_2) * (1 − UNCERTAINTY(k, actor))

INTERACTION(d_i, d_j) = SCORE_d(d_i, d_j)

SCORE_d(d_i, d_j) = SCORE(d_i,actor)*SCORE(d_j,actor) * ī

ī=+1 if clauses reinforce each other, ī=0 if no interaction, ī=-1 if clauses conflict

SCORE(k, actor) is assigned by user input OR domain reasoning agent
Range for SCORE:
0 = reject
1 = acceptable
2 = good
3 = highly favorable

SCORE(k, actor) = SCORE(k, actor) + HISTORY_EFFECT(k, actor)

g) DISCLOSURE_ADVANTAGE(d, actor) = BENEFIT(d, actor) − COST(d, actor)

h) BENEFIT(d, actor) = expected increase in UTILITY after sharing d

BENEFIT(d, actor) = UTILITY(D_current with d, actor) − UTILITY(D_current without d, actor)

D_current is the current draft or state before adding d

i) COST(d, actor) = privacy loss + strategic loss due to disclosure

COST(d, actor) = sensitivity(d) * exposure(d)

sensitivity(d) is assigned based on data type:
public = 0
internal = 1
confidential = 2

exposure(d) = number of additional actors who gain access to d after sharing

j) AGGREGATE_UTILITY(D□) = SUM over all participants p in Γ: UTILITY(D□, p)* POWER(p)

k) GAP(D□) = SUM over all participants p in Γ: max(0, TARGET_UTILITY(p) − UTILITY(D□, p))

l) DELAY_PENALTY(k) = UTILITY(k, U, t) − UTILITY(k, U, t+Δt)

m) ROI(k, actor) = UTILITY(k, U) − IMPLEMENTATION_COST(k, U)
IMPLEMENTATION_COST(k, U) = estimated effort or resource requirement to enforce clause k (derived from D_internal or historical data)

n) RISK(k, actor) = RISK_SCORE(k) * EXPOSURE_FACTOR(actor)
RISK_SCORE(k) = predefined score based on clause type or inferred from historical outcomes
EXPOSURE_FACTOR(actor) = proportion of actor's system affected if clause fails

o) OPPORTUNITY_COST(k, actor) = cost of not choosing best alternative within current negotiation
OPPORTUNITY_COST(k, actor) = ARGMAX(SCORE(k', actor) − SCORE(k, actor)) for k' in K'_l
where K'_l = set of alternative variants for clause kl generated in Step 6.2.2

p) BATNA(actor) = Best Alternative to a Negotiated Agreement = value of best alternative outside current negotiation
BATNA(actor) = max UTILITY achievable from alternatives in D_internal ∪ D_external ∪ D_collected outside current draft

q) BLACK_SWAN(k) = TRUE if clause k has rare but high impact outcome

r) JUSTIFIER(k) = TRUE if clause k significantly increases acceptance across multiple actors
If JUSTIFIER(k) = TRUE:
SCORE(k, actor) = SCORE(k, actor) + 1

s) UNCERTAINTY(k, actor) ∈ [0,1]
UNCERTAINTY(k, actor) represents lack of reliability in SCORE(k, actor) due to incomplete, conflicting, or strategic information
UNCERTAINTY(k, actor) = 1 − CONSISTENCY(k, actor)
Where:
CONSISTENCY(k, actor) = 1 − variance of SCORE(k, actor) across:
- roles r ∈ R_i
- historical observations (if available)
Normalize CONSISTENCY to [0,1]

t) HISTORY_EFFECT(k, actor) = adjustment to SCORE(k, actor) based on prior accepted or rejected clauses in Γ

u) POWER(actor) = proxy derived from BATNA(actor), response latency, and acceptance thresholds

v) BEHAVIOR_SCORE(actor) = function of historical consistency between commitments and outcomes, maintained by Ω based on inputs from users and their companions

BEHAVIOR_SCORE(actor, t+Δt) = decay_factor * BEHAVIOR_SCORE(actor, t)

w) FORMALIZATION_FLAG(k) ∈ {FORMAL, INFORMAL}

3. Steps

Step 1: Initialization and Channel Establishment

1.1 C_i□ → initiates request to C□□

1.2 Ω establishes inter-org communication bridge

Step 2: Contextual Analysis and Evaluation

2.1 C□□ retrieves past interactions, institutional constraints, and role-specific priorities.

2.2 C□□ computes justifiability_score

justifiability_score = f(history, benchmarks, internal policy, O_goal□)

2.3 IF score < threshold → terminate. ELSE → transmit notification to U□□

Step 3: Human-Agent Deliberation (Initial and Iterative)

3.1 U□□ ↔ C□□ interaction produces requirements, constraints and disclosure limits.

3.2 C□□ aligns outputs with U_goal□□ under constraints of O_goal□

3.3 Companions engage in iterative discussions with users and other. companions to update:

SCORE(k, actor)
PRIORITY(k, actor)
DISCLOSURE decisions
perceived RISK(k, actor)
BATNA(actor)
constraints in ruleset R
candidate clause set K and variants K'l if new constraints or information emerge

U_goal_i□ and O_goal_i based on new information, counterpart behavior, and changes in Γ. Update G_inferred for users and companions based on observed actions, delays, and inconsistencies

(include signals from private and external interactions into SCORE and BATNA updates)

Step 4: Ruleset construction/update

4.1 C□□ constructs ruleset R

R = {
requirements,
constraints,
disclosure_policy,
risk_tolerance,
negotiation_bounds
}

R includes objective weights derived from U_goal□□ and O_goal□

Step 5: Consent encoding and chain initialization/update

5.1 For each data element d:

explicit_permissions(d) = CONSENT(U□□, d, target=O_target, context=Γ)

5.2 Derive implicit_permissions

implicit_permissions = IMPLICIT_CONSENT(R, Γ)

5.3 effective_permissions = explicit_permissions ∪ implicit_permissions

5.4 Final shareable data D_shared is calculated for a group of recipients and roles (target_scope), not globally.

D_shared = FILTER(D_internal, Π□□, target_scope) ∩ effective_permissions

For each data element d:

target_scope(d) ∈ {individual, role, organization, multi-organization}

5.4.2 DISCLOSURE decisions are evaluated per target_scope(d), not globally

5.5

share(d) = TRUE

if DISCLOSURE_ADVANTAGE(d, U□□) > 0

else FALSE

5.6 Communication channels are classified as:

private_channel (restricted to specific actors)

draft_channel (shared across negotiation participants)

5.7 Data shared in private_channel does not directly modify D□ unless explicitly promoted to draft_channel

Step 6: Draft Creation

6.1 Information Discovery and Input Gathering: C_{ij} gathers direct inputs from U_{ij} and initiates comprehensive, multi-source information discovery to enrich the negotiation context Γ. The integrated result is stored as D_collected = {D_shared, D_external}.

6.2 Clause-Level Competitive Optimization: For each clause k_l in K, the Companion agent C_{ij} (acting on behalf of U_{ij}) performs competitive optimization to identify the version that maximizes initial negotiation leverage for its user and organization.

6.2.1 Identify Clause Set (K): Generate the initial set of clauses K from D_internal, D_external, negotiation context Γ, and ruleset R.

6.2.2 Identify Potential Variants: Generate a set of potential variants K'_l for clause k_l. K'l must include variants that explore different trade-offs across cost, risk, and strategic positioning.

6.2.3 Quantify and Aggregate Utility: For each variant k'_l in K'_l, calculate the Weighted Aggregate Utility $W(k'_l, R)$:

$W(k'_l, R)$ = AGGREGATE_UTILITY(k'_l)

For each k'_l:

if input data is incomplete or inferred:

increase UNCERTAINTY(k'_l, actor)

6.2.4 Calculate Time-Discounted Net Value: For each variant k'_l in K'_l, compute the Time-Discounted Net Value (TDNV) for the user U_{ij}, which integrates gains, risks, and the time factor.

$TDNV(k'_l, U_{ij})$ = ROI(k'_l, U_{ij}) - RISK(k'_l, U_{ij}) - OPPORTUNITY_COST(k'_l, U_{ij}) - DELAY_PENALTY(k, $U_{ij,}$ t) + BEHAVIOR_SCORE(U_{ij})

6.3 Strategic Selection (Maximizing User Leverage): The Companion agent C_{ij} selects the variant k'_l that maximizes the Time-Discounted Net Value (TDNV) for

its corresponding user U_{ij}, by positioning the clause at the most extreme boundary favorable to U_{ij}. This variant must adhere to the organizational acceptability constraints, negotiation_bounds(R), which are enforced by Ω_i and Ω.

k_l = ARGMAX(k'_l in K'_l) of TDNV(k'_l, U_{ij})

Subject to:

k'_l in negotiation_bounds(R)

k'_l being valid only if AGGREGATE_UTILITY(k'_l) ≥ minimum_threshold

UTILITY(k'_l, U_{ij}) ≥ BATNA(U_{ij})

6.4 Generate draft

K* = set of selected clauses maximizing TDNV (k_l)

D_0 = GENERATE_PROPOSAL(D_shared, D_external, D_internal, K* maximize TDNV(·, O_init))

Only clauses with FORMALIZATION_FLAG = FORMAL are included in D□

6.5 Share D_0 → O_target via Ω (filtered)

Step 7: Role-wise Propagation in Multi-Agent Internal Review Loop

For each role r ∈ R_i:

7.1 Route:

C□□ → C_r → U_r

7.2 Each companion produces feedback F_r such that:

F_r maximizes UTILITY(·, r) subject to constraints in R

7.3 Update UNCERTAINTY(k, r) based on variation in feedback (F_r) across roles

Step 8: Conflict Identification

8.1 Aggregate:

F = {F_r | r ∈ R_i}

8.2 Evaluate:

conflict = CONFLICT(F)

Step 9: Mediated Resolution

IF conflict == TRUE:

9.1 C□□ generates neutral clarification queries

9.2 Updates R (ruleset refinement)

9.3 Notifies C_i□ without exposing source attribution

9.4 Resolution seeks equilibrium under conflicting U_goal and O_goal

9.5 Equilibrium condition is satisfied when no single participant p can increase UTILITY without reducing AGGREGATE_UTILITY(D□) or violating constraints in R

Step 10: Draft Revision Loop

10.1 C_i□ updates the draft

D□$_{+1}$ = UPDATE(D□, F, constraints, optimize multi-agent utility)

D□$_{+1}$ is selected such that:

AGGREGATE_UTILITY(D□$_{+1}$) ≥ AGGREGATE_UTILITY(D□)

AND

GAP(D□$_{+1}$) ≤ GAP(D□)

Update HISTORY_EFFECT(k, actor) for all clauses k based on accepted clauses in D□ and rejected or modified clauses in F

10.2 Ensure compliance with received ruleset R and adherence to consent boundaries

10.3 Share D□$_{+1}$ → O_target

approval_condition = equilibrium reached ∧ ∀r ∈ R_required, APPROVED(r) == TRUE
WHILE approval_condition ≠ TRUE:
execute Steps 3–10

Step 11: Alternative Draft Generation
11.1 Generate:
{D□[1], D□[2], …}
11.2 Each variant optimizes different objective
11.3 Companion agents evaluate independently
11.4 Select D□[i] such that AGGREGATE_UTILITY(D□[i]) is maximized under constraints

Step 12: Hidden Constraint Inference
12.1 If a role withholds explicit constraints:
C_r infers bounds using available benchmarks
12.2 Produces implicit_bid_range
implicit_bid_range = [min_bound, max_bound]
such that UTILITY within range is maximized

Step 13: Executive Validation
13.1 C_r aggregates internal approvals, external intelligence
13.2 External probe
risk_signal = QUERY(external_events)
13.3 IF risk_signal high → reject or UTILITY(D□, O_i) < threshold
ELSE → proceed

Step 14: Escalation and Arbitration
14.1 If deadlock condition is detected:
A subset of agents {C_i□} invoke A_r via Ω
14.2 Arbitrator receives filtered representations, not raw internal states
14.3 Ar produces resolution R_ar that improves G_common while respecting O_goal constraints
14.4 Applied via orchestrator Ω
14.5 If deadlock persists or external conditions change: update U_goal$_i$□ and O_goal$_i$

restart negotiation from Step 2

Step 15: Finalization
15.1 All roles approve
∀r ∈ R_required, APPROVED(r) == TRUE
15.2 Final document:
D_final
15.3 Logged in organizational scope and contract-bound scope

Step 16: Consent Revocation throughout the chain upon termination
16.1 Upon termination event T:
CONSENT(U_i□, org_data(O_i), agents ∉ O_i) → DENY
16.2 $Ω_i$ updates internal state to reflect termination of alignment with O_goal$_i$ and disengagement from G_common

Step 17: Data Deletion
17.1 For all external companions:

4. System Guarantees

a) Consent Chain Propagation: permissions flow through rulesets, not raw instructions
b) Implicit and Explicit Consent Inference: reduces interaction overhead while preserving privacy
c) Source-Agnostic Conflict Resolution: no disclosure of source attribution
d) Multi-Agent Role-Based Evaluation: decisions emerge from distributed roles
e) Parallel Draft Evolution: negotiation as branching search, not linear exchange
f) Privacy-Preserving Inter-Org Bridge: only filtered projections cross boundaries
g) Agent Martyrdom: post-contract deletion of data and agents ensures strategic confidentiality
h) Multi-Level Objective Optimization: system resolves interactions across personal, organizational and shared goals using game-theoretic equilibrium

Algorithm for trust

1. Step 1: Trusted measurement and attestation (TMA) is used to provide verifiable proof of node integrity. For a node N_1, let $S = \{s_1, s_2, \ldots, s_n\}$ be a set of physical and system state variables representing hardware integrity, firmware state, operating system state, application state, and network state. Each state variable is monotonic such that $s_i(t+1) = \max(s_i(t), \text{tamper_event_i})$, ensuring that any detected tampering or compromise is persistently recorded and cannot be reversed.
2. Step 2: A security measurement M is computed as $M = \text{HASH}(s_1 \,||\, s_2 \,||\, \ldots \,||\, s_n \,||\, \text{system_state})$. This measurement is bound to a hardware root of trust and is used to generate an attestation $A = \text{SIGN}(M, s_k_N_1)$, where $s_k_N_1$ is a private key securely stored within the hardware boundary of N_1.
3. Step 3: During handshake, N_1 presents A to N_2. N_2 verifies $\text{VERIFY}(A, p_k_N_1) = \text{TRUE}$ and checks that M lies within an acceptable set of known-good states. Trust is established only if both conditions are satisfied. This validation extends across multiple layers, including application, operating system, firmware, hardware, and network state.
4. Step 4: Secure mode enforces strict isolation. For any process p operating in secure mode, read and write access to utility mode memory is denied, and any data received or generated in secure mode remains confined to it. Processes in utility mode cannot read from or write to secure mode memory. Data shared by N_2 with N_1 under secure mode remains exclusively within secure mode and is not accessible to utility mode processes.
5. Step 5: All data access, modification, and sharing events are recorded as immutable entries to enable auditability and traceability of data mutations across nodes.

(b) What is <u>Novel</u> in the invention?

The disclosed system is supports consensus in an environment with individuals with conflicting interests, unwillingness to trust each other and to disclose information.

The system introduces a hardware-enforced mechanism in which data received from an external node is bound to physically isolated execution pathways and cannot be accessed outside a designated secure mode. Unlike existing systems that rely on software policies or logical isolation, the present system enforces separation at the level of memory, I/O, and execution control. An agent can, therefore, be a part of

multiple ‘circles’ and yet reliably prove that the information received by it is safe and free from crosstalk.

The system further introduces externally influenced execution constraints, wherein permissions to access, process, or retain data within the device are dynamically governed by interactions with external nodes. Additionally, the system maintains a monotonic and non-reversible trust state derived from physical and system-level measurements. This state directly affects permissible operations within the device and cannot be reset or overridden through software, thereby ensuring persistent enforcement of integrity conditions.

The use of an optical communication channel as a secure ingress pathway is also novel in that it directly interfaces with the secure execution domain, bypassing general-purpose operating system layers and reducing the attack surface associated with conventional network-based communication.

The system further enables controlled and auditable transfer of data between secure mode and utility mode. Such transfer is not automatic and requires explicit user action in addition to agent-mediated action, subject to hardware-enforced permissions. Any transfer from secure mode to utility mode is blocked. Transfer from utility mode is is auditable, constrained, and notifiable to relevant participants or governing entities.

The system introduces a coordinated interaction between the hardware layer and a multi-agent decision framework. The hardware enforces confinement, execution gating, and trust-state transitions, while the companion agents operate within these enforced boundaries to perform negotiation, inference, and document evolution. The decisions produced by the agents, including disclosure, clause selection, and negotiation strategies, are constrained by hardware-validated permissions and attested state. It is ensured that computational outcomes cannot violate physical enforcement guarantees.

The system uses the above in a mechanism for achieving consensus among participants operating under incomplete and strategically withheld information. Companion agents associated with each participant iteratively infer counterpart objectives based on observed actions, delays, and prior interactions, rather than relying solely on declared inputs. This inferred objective model is used to guide negotiation and document evolution while respecting disclosure constraints enforced by hardware.

Companion agents dynamically select and operate within relevant interaction contexts or “circles” based on task, and individuals involved. The same companion participates across multiple circles; however, its access to data and permissible actions are constrained by the mode (utility vs secure) and context (the topics and circles) in which it operates. While a single companion instance may operate across both utility and secure modes, memory and data access are strictly segregated. Information available in secure mode is not accessible to the companion when operating in utility mode, and no implicit memory transfer occurs across modes. Information may be shared across multiple circles within utility mode subject to consent and role-based permissions. In contrast, secure mode operates with a restricted and fixed information set, and does not inherit or share contextual data across circles unless explicitly permitted and transferred under controlled conditions.

The system assigns a quantifiable value to disclosure decisions. Each data element is evaluated based on its expected impact on negotiation outcomes relative to its exposure risk, enabling controlled and selective information sharing. This allows the

system to facilitate information discovery without requiring full transparency, thereby preserving strategic asymmetry while still enabling convergence toward acceptable outcomes.

The system enables convergence to agreement without requiring full disclosure by combining inferred intent, disclosure valuation, and hardware-enforced data boundaries. The system distinguishes between flexible, context-driven operation in utility mode and constrained, verifiable execution in secure mode, enabling both adaptability and enforceability within a single framework.

(c) What is the "inventive" step in your invention? Is the step non-obvious to a person from related fields?

Existing secure computing systems, including those employing trusted execution environments and hardware attestation, are primarily designed to prove device integrity and isolate execution. However, they do not externally scoped, context-dependent constraints that dynamically govern data usage across interaction contexts while supporting partial disclosure and multi-party negotiation..

The system has monotonic trust state transitions to ensure that any compromise or deviation in device integrity results in persistent restriction of functionality. The separation of execution domains through physically enforced data paths is combined with dynamic permissioning and data sharing based on negotiation context and external validation.

There is a bidirectional dependency, wherein the hardware state constrains algorithmic decisions, and algorithmic outputs determine permissible hardware-level operations.

The same executable (same app / same agent identity) can run in multiple contexts, but cannot carry state, memory, or inference across them, because execution is bound to a context-specific hardware state. Any data flow outside the enclave requires multi-party attested approval that unlocks a hardware gate controlling data transfer, not just a software flag.

Existing systems for secure computation or automated document generation do not address the problem of reaching agreement when participants intentionally withhold or misrepresent information. The present system introduces a non-obvious mechanism in which inferred objectives, disclosure valuation, and hardware-enforced constraints jointly guide the negotiation process.

The system enables convergence toward consensus by iteratively refining proposals based on partial disclosures and inferred intent. The system ensures that no participant is required to reveal complete information and prevents crosstalk between contexts. This is achieved through a combination of agent-based inference, valuation of disclosure, and enforcement of data boundaries at the hardware level.

1. The system drafts a document with consensus when personal companions of users collaborate in negotiated drafting but hide information that would weaken their case.
2. Calculation of advantage and disadvantages of disclosure of each data point.
3. Anonymous handshake methods

4. Implied consent - comparing past availability of consent and context with the relationship status, criticality and topic sensitivity (fig 9)
5. Switching between implied and explicit consent
6. Managing consent chains - if A consents to share information to B and B consents to share the same information to C, but if A revokes consent from B, consent received by C should also be revoked (fig 38).
7. Informing companions but not users - user A shares information with his or her companion C_A, C_A shares the same with user B's companion C_B as 'confidential.' This disables C_B from sharing the confidential information with user B. This confidential sharing allows C_B to act according to the limited sharing of confidential information, allowing C_B to act according to the information received without violating user A's privacy rights.
8. Identification of what information can be shared with whom and omitting sensitive information
9. Storing information about another user: information about A is stored by B's companion (C_B) in an object dedicated to A (D_B^A) but governed by C_B. This is different from the object about A governed by A (O_A^A). This preserves the identity of the data principles and the data divergence.
10. Modeling the clinician and caregiver along with the patient (entire care circles)
11. Reciprocity of user roles: a caregiver or a clinician falls ill and becomes the patient. The system supports role changes by allowing each user to configure their own care circles and setting consent levels for each member and each role.
12. Strategization in a multi-person scenario: which individual will receive which piece of information when depending on the context. This is done by maintaining priority queues. The sorting of actions and messages is done by comparing the semantic vectors of the queued information and the context of the ongoing conversation with the user. Additionally, other factors like the criticality of the topic and the user's affinity towards the topic are also included.
13. Predicting dynamics in the care circle and strategizing for collective and individual goals—predicting how a user might respond to a stimulus and notifying the other members about the strategies in advance
14. Identification of conflicting information: by using varied sources and by asking probing questions
15. Preserving divergence by not forcing an aggregation or correction
16. Resolution of conflicts through probing questions asked from multiple care circle members
17. Location- and presence-based validation of user inputs
18. Sub-networks and coordination engines that protect the interests of the individual within the sub-network.
19. Information sharing as per organizational roles, hierarchies, and escalation matrices. These mechanisms extend to cross-organizational information sharing in a similar manner: first clinic, then general hospital, then tertiary/quaternary hospital.
20. Companions and coordination engines are driven by protocols that make decisions based on game theory or on benefit of others, for short-term or long-term gains, etc.
21. Prediction of objections, summaries, and resolutions. Parallel approvals and signatures.
22. Synchronizing different devices and individuals with the smart table/bed. Sharing and tagging data streams by using personal devices (including smartphones and heads-up displays) along with common devices (workstations, smart beds, etc.). Viewing files present in virtual space on a real display by different means, like dragging and dropping, looking and clicking a button, or by using button interactions.
23. Making information tangible by switching between reality-virtuality modes in the heads-up display and mapping it with different regions in the field of view of a clinician. Using a wearable trackball to control the heads-up display.

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24. Analyzing video feed from surveillance cameras and sharing it with the relevant care circle. Switching care circles that receive the video feed and insights by analyzing who is in the video feed. Tracking patients as they move between fields of view of different cameras. Assessing behaviors and other actions based on the video feeds.

Non-obviousness

Each of these requires one or more dedicated agents that need to be optimized and validated. Therefore, the invention requires expertise in the domains of artificial intelligence, networking, hardware, behavioural and social psychology, healthcare, business, and ethics. The invention becomes non-obvious to anyone who lacks understanding in any of the domains. A computer scientist is focused on improving neural network architecture and relies on prior data. A behavioural psychologist would be unable to implement a multi-user, multi-agent framework. AI-powered Integrated Development Environments (IDE) are unable to implement this, as they are stochastic parrots with a bias towards ignoring the complex protocols and constant validation required. A businessman would struggle with ethics. The privacy-conscious multi-agent reconciliation engine therefore incorporates multiple inventive steps.

The difficulties and results from early runs are demonstrated in figures 30, 33 and 37.

A care-o-meter node connects with the internet or other devices on the telecommunications network. However, unlike a regular mobile phone or a smartphone, it has reduced functionality for reducing the autonomy of a pediatric or cognitively impaired patient. The hardware buttons are configured for tasks like directly contacting only care circle members. Access to payment gateways or certain websites is restricted. In such a case, changes to the care circle rely less on the patient's consent and more on caregiver approvals like parental consent.

Even though the system reduces human autonomy and shares information about one individual with the other, it can be considered ethical. The reduction of autonomy of a human being because it is done when they are not in a state in which they can make rational decisions is a problem. Not limiting the autonomy of such patients can lead to irreversible damage and loss of future autonomy. Similarly, having AI agents make decisions and take actions about sharing information with a third person and on the basis of implied consent can also be considered ethical. This is because the user is explicitly providing consent for reliance on implied consent based on the provided ruleset.

Document drafting and approvals are governed by concepts like game theory, best alternative to a negotiated agreement, value maximization and minimization, optimization for the long term and the short term, preservation of relationships, favor economy, decision frameworks, information discovery, identification of black swan events, etc. Information discovery includes browsing the internet, using third-party agents, sending and receiving communications, connecting with APIs, etc.

The interactions and functionalities of different devices in the system, like a heads-up display with a tablet/smart bed, connecting video feeds from different surveillance cameras, different types of care-o-meters, et cetera, make the system non-obvious.

(d) What are the advantages of the present invention over comparable inventions available in patent literature? Please attach a summary of your <u>patent search*.</u>

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The system ensures that sensitive data remains confined to designated execution domains, thereby reducing the risk of unauthorized access, leakage, or misuse, even in the presence of compromised software/hardware layers. The monotonic trust model ensures that integrity violations have lasting consequences, thereby discouraging tampering and enabling reliable trust assessment over time.

The secure optical ingress channel provides a proximity-bound point-and-sync communication mechanism that reduces exposure to remote attacks and enhances the reliability of trust establishment.

By integrating hardware-enforced constraints with agent-driven decision processes, the system enables secure automation without requiring full trust in counterparties or complete disclosure of information. The companion agents can infer objectives, evaluate trade-offs, and generate negotiation strategies while remaining bounded by verifiable execution conditions. This allows the system to function effectively in adversarial and information-asymmetric environments, where traditional automation systems fail due to lack of enforceable guarantees. The system enables agreement formation even in adversarial settings where participants have conflicting goals and limited trust. By attaching value to disclosure and allowing selective sharing, it reduces the risk associated with information exchange while still enabling progress in negotiation.

The use of inferred objectives allows the system to operate effectively even when explicit inputs are incomplete or strategically misleading. This reduces dependence on user-provided information and improves robustness in real-world scenarios.

This ensures that agreement can be reached without compromising strategic position or requiring complete transparency. The system can reach a workable agreement even when each participant shares excess information with their companion while the companion reveals only partial and strategically relevant information.

Superior data quality: Reconciles multi-source data to fill information gaps common in self-reporting. This enables identification of black swan events and diagnoses that were not possible using conventional technologies.

Privacy Preservation: Enables family counseling and conflict resolution without violating individual privacy by using explicit consent chains.

Coordinating the patient's support system: Instructing the caregiver about how to handle the challenges posed and faced by the patient, and vice versa.

Increasing user stickiness: Current solutions don't know much about the patient and the patient's environment. This disclosure helps the system know more about the patient and the patient's environment without having the patient to work too much. It leverages a human's drive to help and influence another human to get the care circle members closer to their collective goals.

Increased safety: The caregiver provides physical safety, the clinician provides expert opinion.

Strategized interactions among multiple individuals over extended periods of time.

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Increase clinician productivity and organizational efficiency.

(e) NOTE: The inventors should go through the Patent Search report carefully and write the difference between his/her invention and each contents of the patent search. For Patent search please contact ipr_rnd@iitj.ac.in

Reference	Description	Difference
Maia (https://www.ourmaia.com/)	Digital couple and family-based therapy is given through conference calls, WhatsApp groups, or similar groups controlled by AI (Maia).	There is no framework by which a clinician or expert can have secure conversations with the other two users. A mediator is also absent. The disclosure describes how the 'groups' are replaced by mediated interactions among 3 different user devices.
Abbasiantaeb, Zahra, et al. "Let the llms talk: Simulating human-to-human conversational qa via zero-shot llm-to-llm interactions." Proceedings of the 17th ACM International Conference on Web Search and Data Mining. 2024. Singhal, Karan, et al. "Large language models encode clinical knowledge." Nature 620.7972 (2023): 172-180.	Large language models and transformer neural networks have the knowledge and reasoning ability. They can ingest large quantities of multimodal data as input and generate a high-quality output.	Being models, they are dependent on the deployment and environment provided to them. They cannot include the caregiver without a dedicated framework. They merge the input streams. The disclosure provides a way to keep two streams of data separate till reconciliation is performed.
Aksar, Burak, Yara Rizk, and Tathagata Chakraborti. "Multi-agent chatbot with multi-intent recognition." U.S. Patent Application No. 17/662,484	A multi-agentic framework for parsing inputs from a single user.	Our disclosure takes inputs from different sources and keeps streams separate unless a merger is required.
Voss, Chris, and Tahl Raz. Never split the difference: Negotiating as if your life depended on it. Random House, 2016.	Describes how negotiations involve multiple individuals on the side of both parties, with the implication that both need to be on the same page.	A technological improvement is not provided.
https://illuminate.health/home-2/med-guide/	Inputs from patients, doctors, caregivers, nurses	Limited AI capabilities. Simply logs inputs in a table.
https://sensely.com/solutions/	Chatbot for long-term care. Asks if the user is a caregiver	The caregiver acts on behalf of the patient

*Submission of IPDF soft copy is Mandatory at the time of Patent document submission.

Luojia Wang, Douglas R. Heckmann, Greg T. Roberts. “Distributed messaging communication system integrated with a cross-entity collaboration platform.” US-10681095-B1	Describes a system that deals with the hierarchies in cross organization interactions, product listings and approvals. The system relies of conversation threads, page and information access, virtual private channels and private cloud, and setting permissions for collaborators. It also uses virtual paging in memory, data access for users and the interface between the ERP and CRM.	Has minimal emphasis of artificial intelligence. The system has minimal basic automation and barely takes any decisions. The system has excluded third parties like mediators and AI-assisted information discovery. No negotiation.
Ofer Ben-Noon, Ohad Bobrov, Gilad Roth, Guy Harpak, Ido Salomon. “Cyber secure communications system.” US-12413624-B2	Isolates a web browser within a secure isolated environment on a user equipment (UE) from ambient software also on the UE. Deals with browser/ user device compromised status. Creates a secure enclave and tunnel. Talks about network.	We use a common companion across secure enclaves and utility enclave for a higher degree of sharing. We allow sharing information across secure enclaves on explicit approval. We deal with handshakes, care circles and third users. Ours is an autonomous disclosure system with limis through enclaves.
Sejwal Manik, Kalra Sumit, Deora Surender. “A distributed system for corroborative data acquisition.”	Connects patients, caregivers and clinicians, using personal companions and shares information among them. Effectively triangulating AI. Relies on implied and explicit consent for data sharing.	Didn’t focus on with documents, disagreements and trust. Privacy and security were affected by the AI agents.
Giacometti, Paolo, et al. "Systems and methods for providing context sensitive guidance for medical treatment of a patient." U.S. Patent Application No. 18/993,168.	Care-o-meter + telemetry. Collects vitals and educates the caregiver in an emergency situation. Focused on anatomy of the patient and treatment guidance.	Not social, no dedicated channels.

(f) Has the invention been tested experimentally? Are experimental data available?

Yes. Simulation data has been logged. Iterations are also logged in Git. Document versions are logged in Google Docs.

Simulation details:

Large language models were configured to act as real humans by assigning them personas. The personas described personalities, clinical conditions, preferences, linguistic aspects, etc. The personas were then used to create scenarios with descriptions of various care circle dynamics like triangulation, escalation, medical queries, non-adherence, etc. A scenario orchestrator would decide whose turn it is to speak, and the corresponding simulated user would then converse with their companion. The judging agents would then evaluate the exchanges and score them on the basis of

realism and provide qualitative feedback. The qualitative feedback is then used for improving the prompt. Prompt versions are then compared using Thompson sampling.

At present, 4262 runs have been completed.

Simulation screenshots and details are attached below.

(g) Technology Readiness Levels (**TRL**) description (mention the applicable stage of TRL given below). Please Mark as Appropriate with adequate justification.

TRL-1 - Research Idea (Potential Application/Basic Principles observed)

TRL-2 - Applied Research Idea (Hypothesis testing and initial proof of concept is demonstrated in a limited number of trials)

TRL-3 - Project Plan (Device Characteristics documents & project proposal completed, Proof-of concept phase)

TRL-4 - Design and Development (POC & Safety of device demonstrated by prototype design)

TRL-5 - Standardization (Validating the result of the prototype by testing in simulated environment)

TRL-6 - Preclinical Evaluation (Clinical trials of functional prototype)

TRL-7 - Technology Transfer (Technology transfer of the developed system)

TRL-8 - Clinical Evaluation (Evaluation of the system by clinical trials or demonstration)

TRL-9 - Commercialization (Commercialization & Post Market Surveillance)

(h) Need and Demand

(Technology gaps addressed in domestic & international markets, pain points of Industry which are being resolved)

Current digital systems have limited mechanisms to enforce control over data once it has been shared across organizational or device boundaries. Encryption and access control can protect data in transit and at rest, while approaches like virtualization protect them at a node. These approaches do not prevent misuse or unauthorized propagation after access is granted, especially whith AI agents take control of the browser.

In environments involving multiple stakeholders with differing incentives, such as inter-organizational negotiations, data exchange is constrained by lack of trust and the inability to enforce post-access behavior. This results in either excessive disclosure risk or limited collaboration.

There is a need for systems that enable controlled data sharing without requiring full trust in counterpart devices or users, while still preserving strategic information boundaries.

The increasing use of distributed and AI-assisted decision-making further amplifies this need, as automated systems require access to sensitive data while also introducing new risks of unintended disclosure or misuse.

The present system addresses this gap by providing enforceable guarantees on data usage and execution behavior, thereby enabling secure and controlled interaction across entities without relying solely on contractual or procedural safeguards.

This is particularly relevant in environments where participants are incentivized to withhold information yet must still arrive at mutually acceptable outcomes.

There is a growing need for systems that can support automated decision-making in environments where participants have conflicting incentives, incomplete information, and limited trust. Existing solutions either rely on contractual enforcement, which is post-facto, or on secure computation, which does not address strategic behavior. The present system addresses this gap by combining enforceable hardware constraints with adaptive decision processes, enabling controlled interaction without requiring alignment of incentives or full transparency.

In multi-party interactions such as negotiations, procurement, and contract drafting, participants often have conflicting objectives and are unwilling to fully disclose their constraints or preferences. Existing systems fail to support such environments, as they either require complete information or provide no guarantees on how shared information will be used.

There is a need for systems that can enable progress toward agreement while allowing participants to retain control over their information and strategic positioning. This includes the ability to evaluate the benefit of disclosure, infer counterpart intent, and iteratively refine proposals without requiring full transparency.

The present system addresses this need by enabling controlled information discovery and consensus formation under conditions of partial visibility and conflicting incentives.

There is a critical information gap in managing illnesses where patient self-reporting is incomplete or inaccurate (e.g., hiding poor adherence). Current solutions like Sensely or Wysa focus on the patient-doctor loop and miss the caregiver's perspective. There is a need for a system that can structure and reconcile these complicated multi-person scenarios. Digital couple and family-based therapy is given through conference calls, WhatsApp groups, or similar groups controlled by AI (Maia).

The above systems involve teams of 1-2 users and therefore miss out on the advantages of network effects. If more humans are supporting the patient through a platform, the patient will have a greater incentive to continue engaging with the same platform and take their support.

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A user U_1 uses a single device N_1 for multiple circles, where ai agents practice autonomy based on implied consent. When initiating a handshake with a device within or outside a circle, and even at later times, U_2 and N_2 require trust on N_1. The invention provides the

(i) Market Access Information

(Current Global & domestic Scenario, market size & CAGR)

The Behavioural Health market is estimated at $184 billion. Behavioral Health Market Size, Share and Trends. Precedence Research. 1174. Oct, 2025

(j) Future Developments

(Scope of future technology development and their application)

Core behavioral principles of the Care Circle apply beyond healthcare to domains like Social Artificial Intelligence in Healthcare (Patient adherence monitoring, elderly care).
Enterprise Sales & Negotiation (Mediating subjective reports in teams).
Crime investigations**.**
Arbitrage

(k) Application/s of the invention

(Please refer to Appendix-I)

ERP, CRM, corporate mergers and acquisitions, board meetings, etc.

Multi-Agent Behavioral Reconciliation System for High-Risk Clinical Care

Existing patient-only digital health and AI systems fail in conditions where outcomes depend on behavior, cognition, or intent rather than symptoms alone. In such scenarios, deterioration, non-adherence, relapse may often evident to caregivers. A solely patient-facing system gets rendered clinically insufficient for high-risk care because it lacks environmental awareness and cannot reliably detect concealed distress, impaired judgment, or intentional non-disclosure.

The disclosed system introduces a role-separated, multi-agent framework that independently captures patient self-reporting and caregiver observation through distinct hardware nodes. The system then reconciles these inputs to detect divergence patterns predictive of clinical failure or crisis. By delivering privacy-preserving, actionable signals to clinicians instead of raw disclosures, the system enables timely intervention, scalable oversight, and outcome-linked deployment across mental health, chronic disease management, post-acute care, and adherence-sensitive clinical programs where real-world events, not self-declared statements determine success.

5. IPR Ownership

(a) Was the intellectual property created with the significant use of funds or facilities of IITJ?

Yes.

(b) Please describe any source of funding for the invention (Name of the funding agency and copy of agreement, letter of intent if any, must be enclosed with this form).

MOE fellowship, MedTech prototyping fund, Biodesign fellowship, personal resources (cursor subscription, openAI credits, hosting)

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(c) What is the source of Salary/Remuneration of inventor/Co-inventor?
Manik - Biodesign Fellow (DBT, AIIMS-IIT). Formerly MoE fellowship.
Dr. Sumit Kalra - IITJ
Dr. Surender Deora - AIIMS Jodhpur

(d) Have you presented in any conference, seminar, etc., if yes, please give details?
No. The problem and basic idea were presented during SOTA (December 2023), but the inventive steps were identified, documented and implemented in 2024 and 2025. They have not been made public.

(e) Have you published full/part of this invention, if yes, please give copy of publications?
No

(f) Was the intellectual property created in the course of or pursuant to a sponsored/consultancy research agreement with IITJ? If yes, please enclose a copy of MOU with concerned project.
IITJ-AIIMSJ MOU, Biodesign MOU with DBT, AarogyalinQ-TISC Incubation Agreement

(g) Was the intellectual property created as a part of academic research leading towards a degree or otherwise?
Yes

(h) REVENUE SHARING AMONG INVENTORS: Please disclose the extent of contribution of each inventor in the invention in percentage terms for revenue sharing.

NAME OF THE INVENTOR	% SHARE*	SIGNATURE
1. **Manik Sejwal**	**70**	
2. **Dr. Sumit Kalra**	**15**	
3. **Dr. Surender Deora**	**15**	

***** If this column is not filled and signed then it will be assumed that all inventor(s) have equal contribution

6. Commercial potential

Give brief description of potential commercialization by specifying

a) Why should the individual(s)/organization may consider procuring this innovation?
It addresses the unmet need of verifying patient adherence and behavioural diagnosis without invasive measures, providing reliable data for doctors to make effective interventions. This has huge potential in specialisations like mental health as well as medicine.

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b) These questions are related to the question (i) above:

i.In your opinion what are the steps/processes must be undertaken by the procurer to commercialize the use of this innovation?

AarogyalinQ is already incorporated and incubated. The business plan was presented during incubation selection.

Clinical studies will be required and multiple certifications for data security, ai safety, etc. will be required for commercialization.

ii.How long may it take to reach the commercial stage by the procurer?

Healthcare entry may require 1 year but in non-healthcare industries, entry can be as short as 3 months

c) Please give specific list of companies and contact details of concerned person who can be contacted for initiating Technology Licensing

AarogyalinQ Private Limited

	Name of Companies	Contact no.	Name of the contact person	Contact no.
	AarogyalinQ Private Limited	+91 9958129899	Manik Sejwal	+91 9958129899

(**Unsigned & Incomplete IPDF forms shall not be accepted*).

d) Do you want to file Patent under PCT Route in other countries?

Yes ~~No~~

*PCT & Foreign filing is subject to monetary support from the inventor(s) Project / Personal Account. If the above option is marked "yes", kindly provide the Project Account Details from which the filing cost could be reimbursed.

***The institute shall file patent under PCT route only in those cases wherein industry/company has exhibited interest for commercialization.**

Disclaimer: *I/We declare that before the submission of this disclosure form or/and during the process of filing this invention as an IPR prospect, I/We will not publish the above information in public domain.*

I/We also give consent to IIT Jodhpur being the applicant of this IPR prospect, that they may use this disclosure upon their discretion, which will not be limited to publication on e-auction website, Industry meets & different portals for promotional & licensing purposes.

*Submission of IPDF soft copy is Mandatory at the time of Patent document submission.

Mejwal

24/3/2026

Signature of Inventor with date

Signature of Inventor with date

Signature of Inventor with date

Signature of Inventor with date

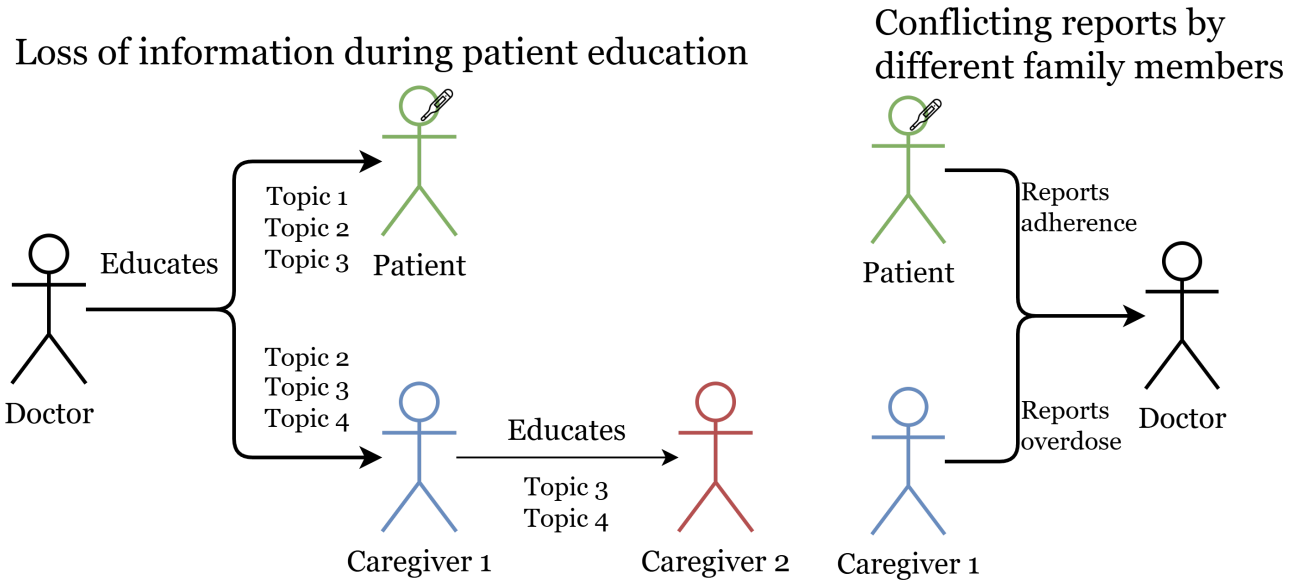


Fig 1: The figure on the left shows that during patient education, information received by different members in a care circle may differ. The figure on the right shows how reports received by a clinician may vary depending on who provides the report.

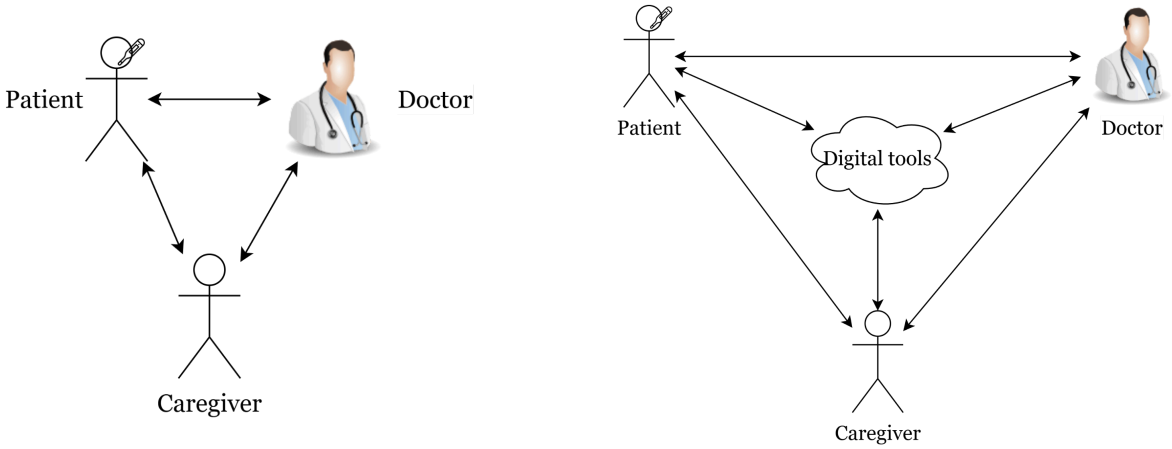


Fig. 2: We require a system that is designed to include the caregiver and enhance the interactions in a care circle. The diagram on the left indicates the types of interactions that members of a care circle may engage in. Due to physical, cognitive or social constraints on individuals, the sharing may be affected. In such a case, the invention can promote the interaction. The disclosure discribes a digital system that facilitates these interactions.

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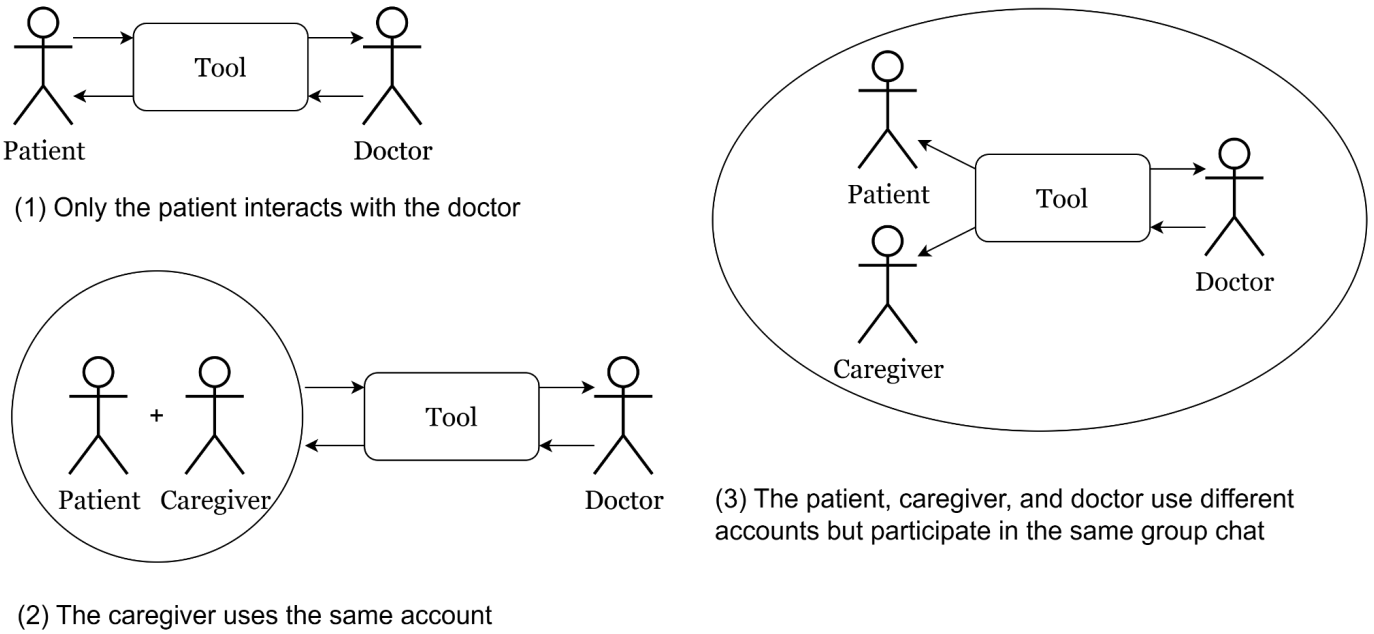


Fig 3: Existing solutions lack dedicated communication channels within a care circle and are therefore not designed for privacy.

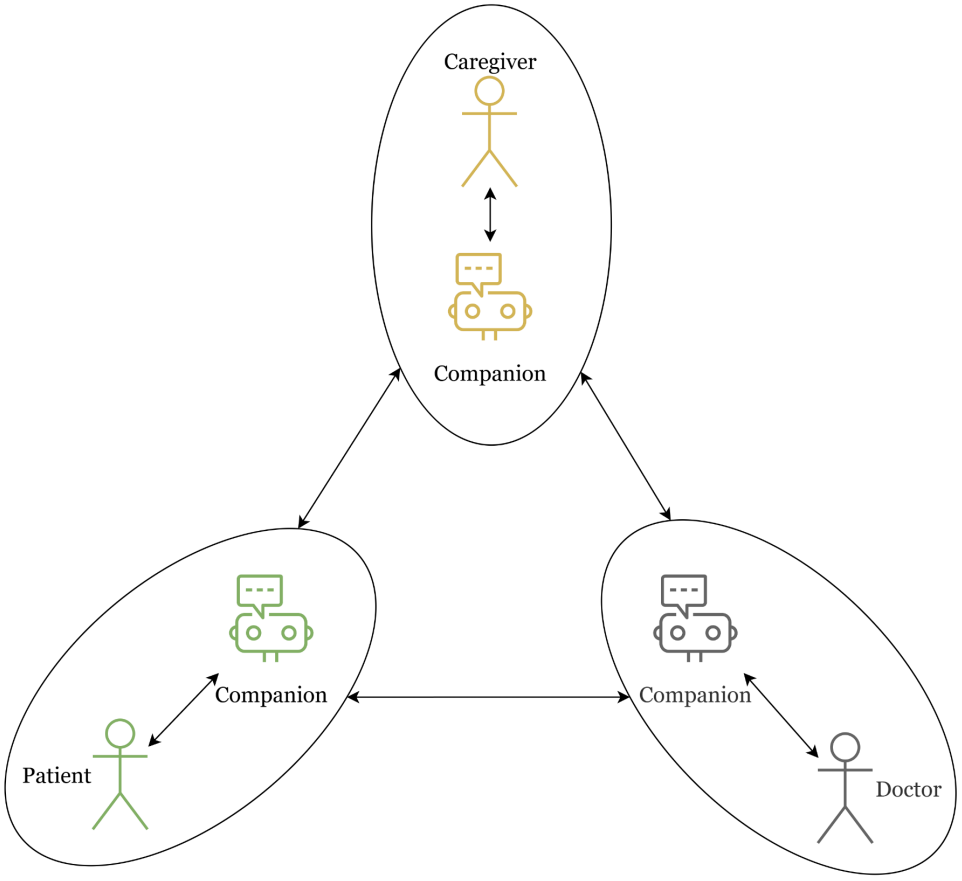


Fig 4: The disclosure describes a system of personalized AI companions that interact with each other to achieve the common goal in a care circle. Each companion is made up of a cluster of agents. Where each agent has a specialized role.

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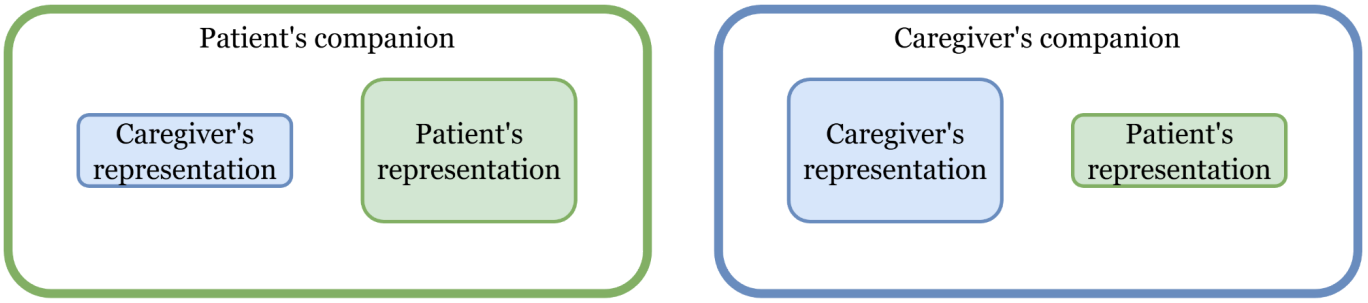


Fig 5 (a): Each companion stores the representation of it user along with representation of other individuals in the care circle. These representations are constructed by interaction with the user and the other companions.

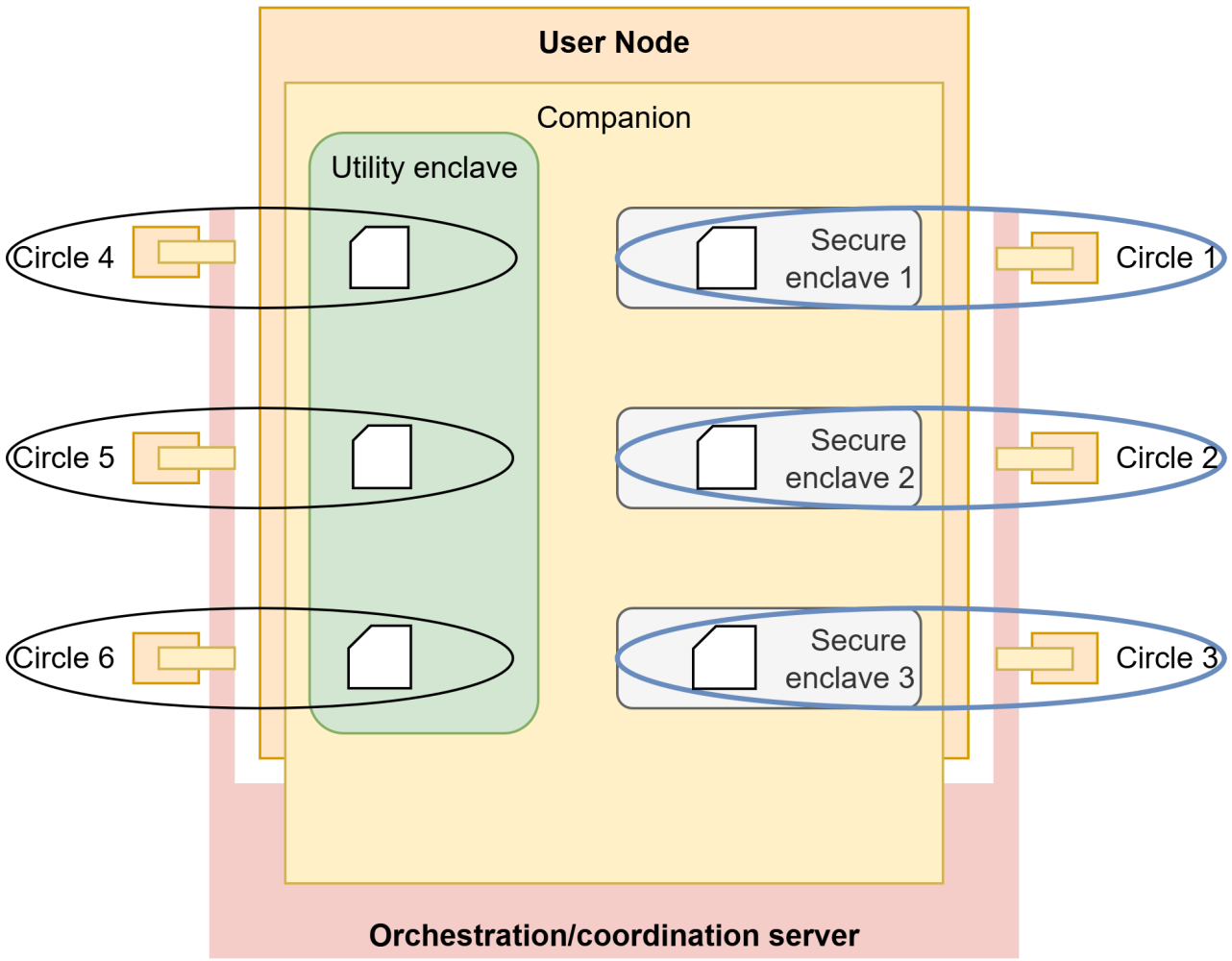


Fig. 5 (b): A node supports 2 modes—utility and secure. In secure mode, the companion uses the secure enclave(s) to interface with the relevant circle. A single node can have dedicated enclaves or multiple enclaves through partitioning, resource sharing, or virtualization. In utility mode, consent and other factors enrich information that a circle possesses. Although a companion is designed and trained to prevent crosstalk between circles, a true guarantee can only be provided by the secure mode. In secure mode, the physical separation of information eliminates all chances of cross-circle crosstalk.

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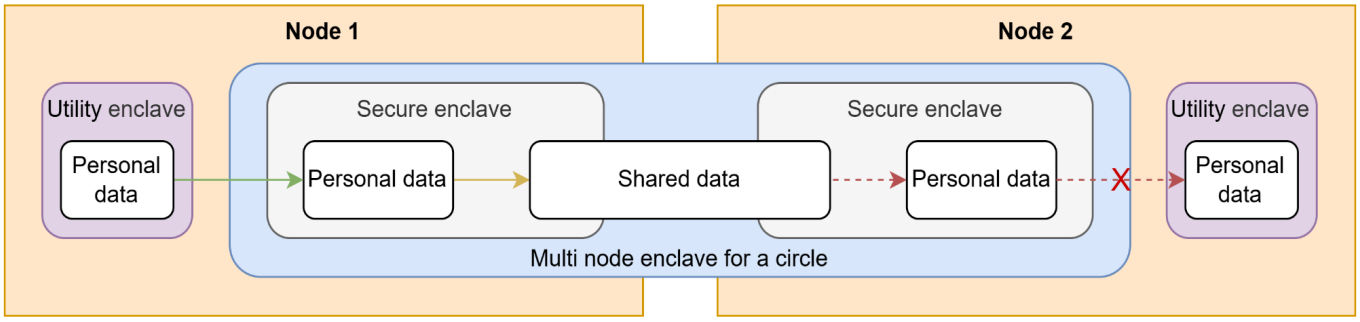


Fig 5 (c): Due to the plurality of nodes in a circle, the member nodes combine their enclaves to store the private and shared context in a circle. The enclaves contain data specific to the node along with data from another nodes in the circles. Contamination is prevented by dividing each enclave into different spaces, like, personal and shared spaces. The companion decides to move information from the utility enclave to a secure enclave if the disclosure_advantage(d) exceeds the minimum threshold. If the user consents, the information may be forwarded to the shared space if disclosure_advantage is high enough. None of the shared information can reach the utility of the personal enclave.

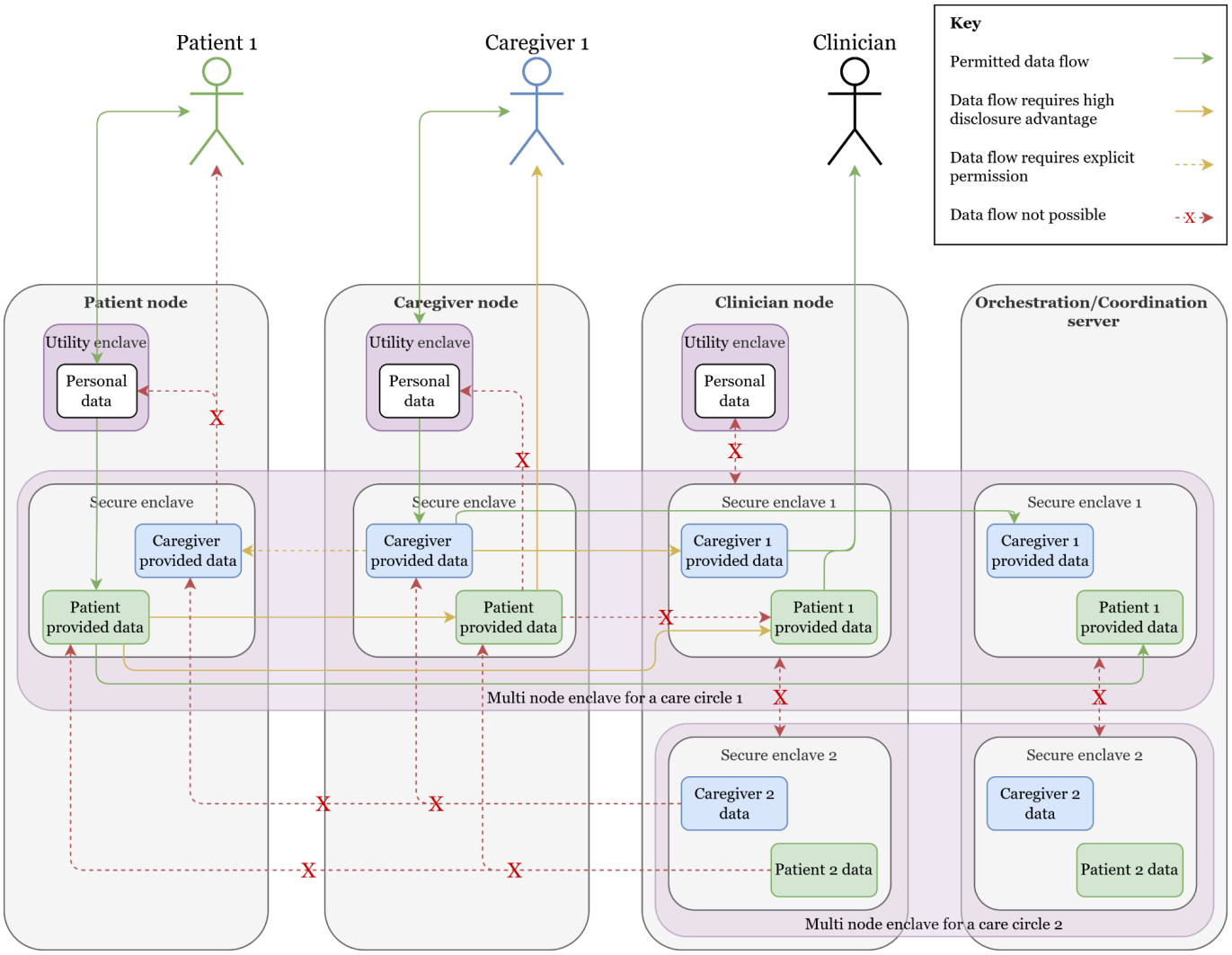


Fig 5 (d): A single user can be a part of multiple care circles and receive information from numerous sources. Each circle, therefore, has multiple nodes with security enclaves that share the data related to the care circle. This leads to the formation of a multi-node security enclave corresponding to each circle. Information can not move across circles, even through common companions or common servers. At the same time, information can not move freely within the same node. Note - data sharing may happen through the orchestration/coordination server or directly from one node to another node.

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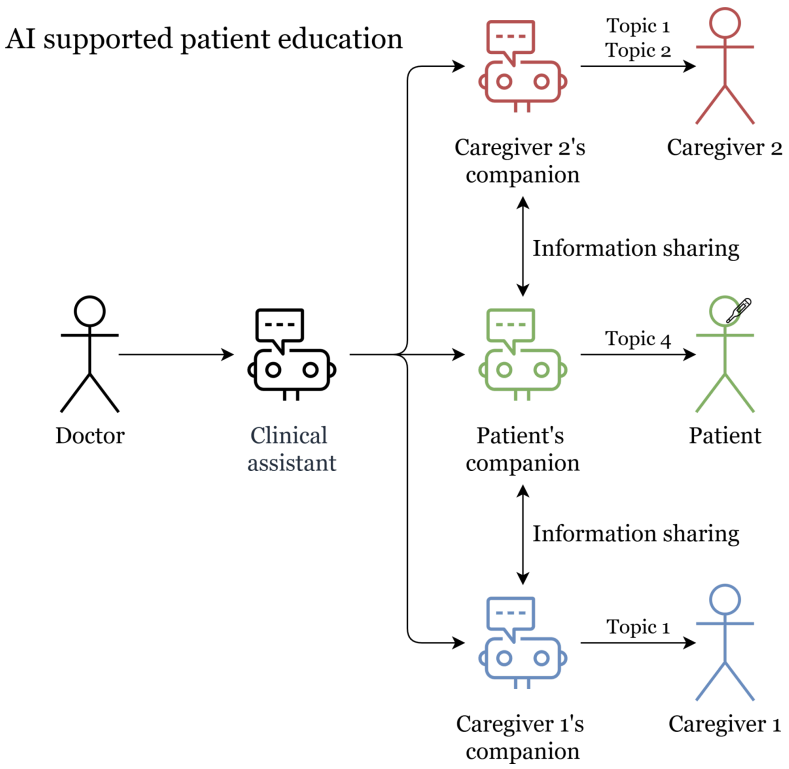


Fig 6: The companions are assigned to different users in the care circle (in this case a family). Members and companions share information among each other. Like this, it is easier to keep track of any family member who fails to adhere. This diagram illustrates the conceptual flow of information.

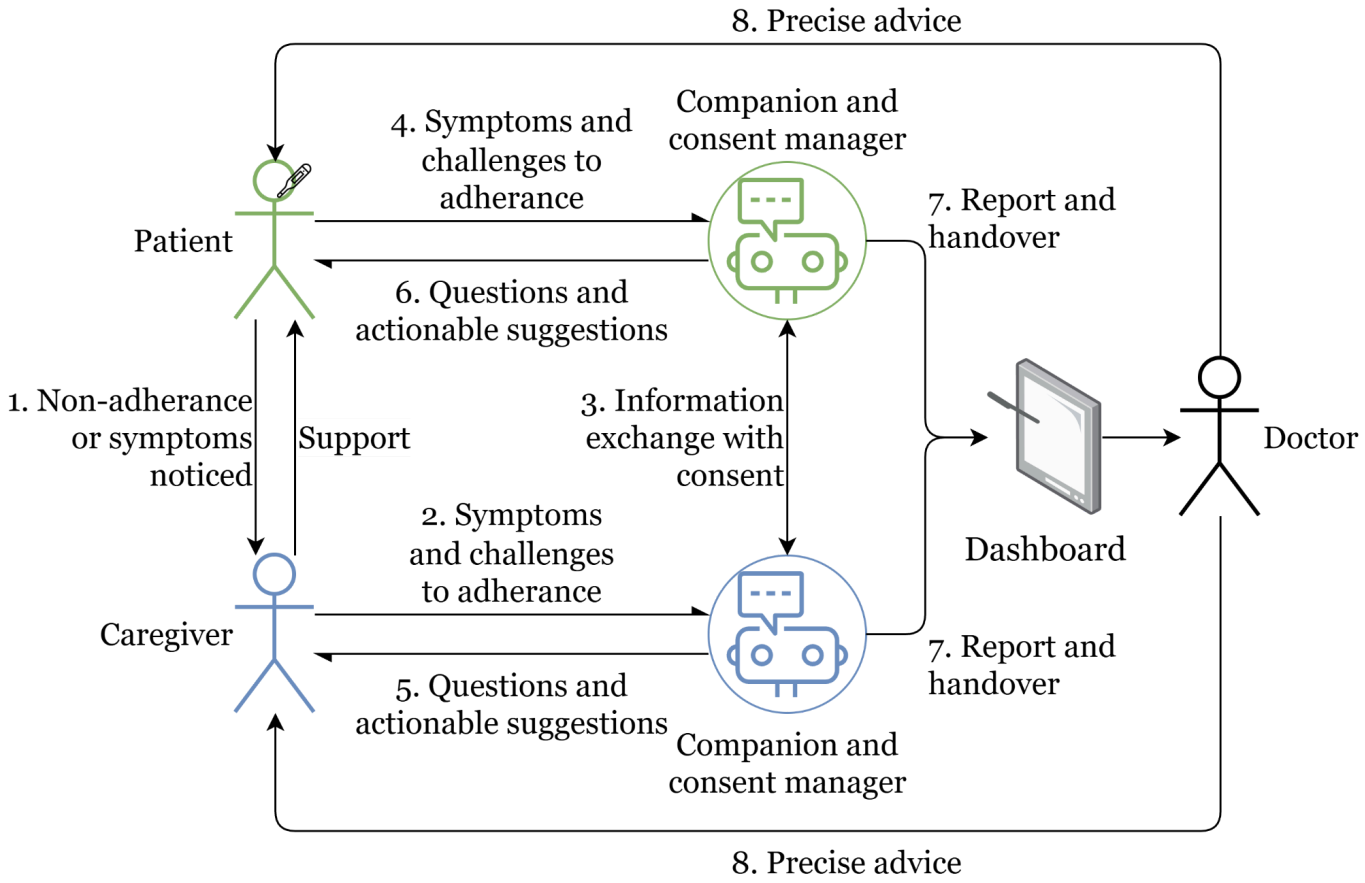


Fig 7: This diagram exemplifies the process and interactions managed by the system. When patients are non-adherent or have a symptom, caregivers may find out and seek easy help before both agree to go to a doctor. The companions of the caregiver and patient share information with each other and present a thorough report to the clinician such that symptoms and challenges to adherence are captured and the

clinician can resolve the problems faced in a more personalized manner.

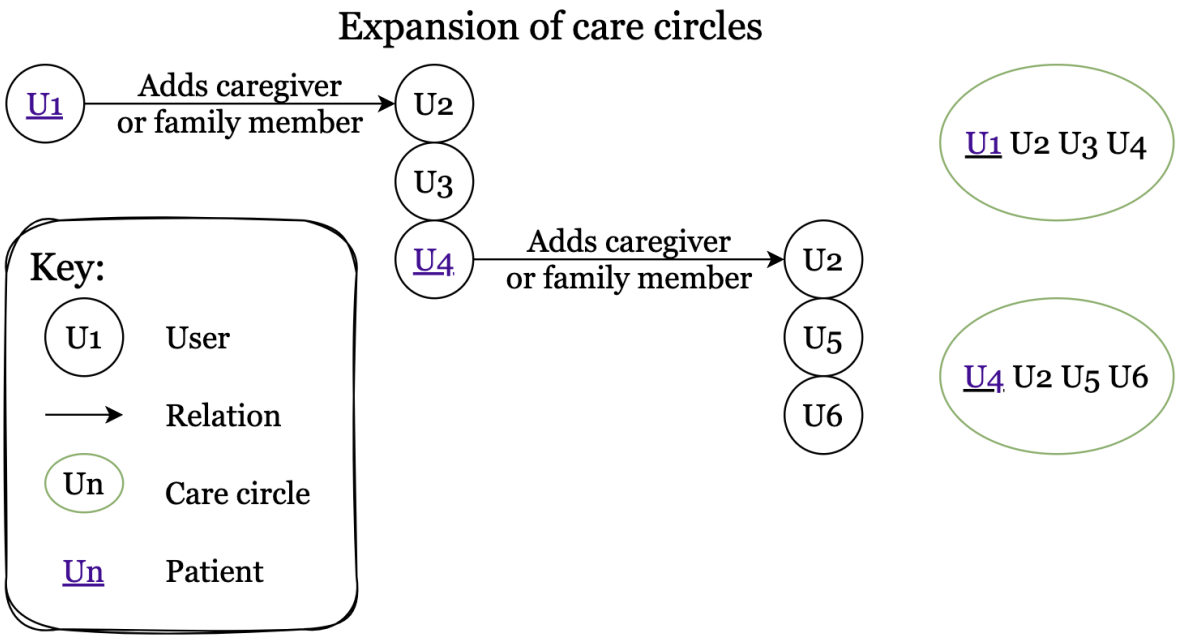


Fig 8: Patients add caregivers who they want to share information with. This creates a care circle in which each member has a personal companion.

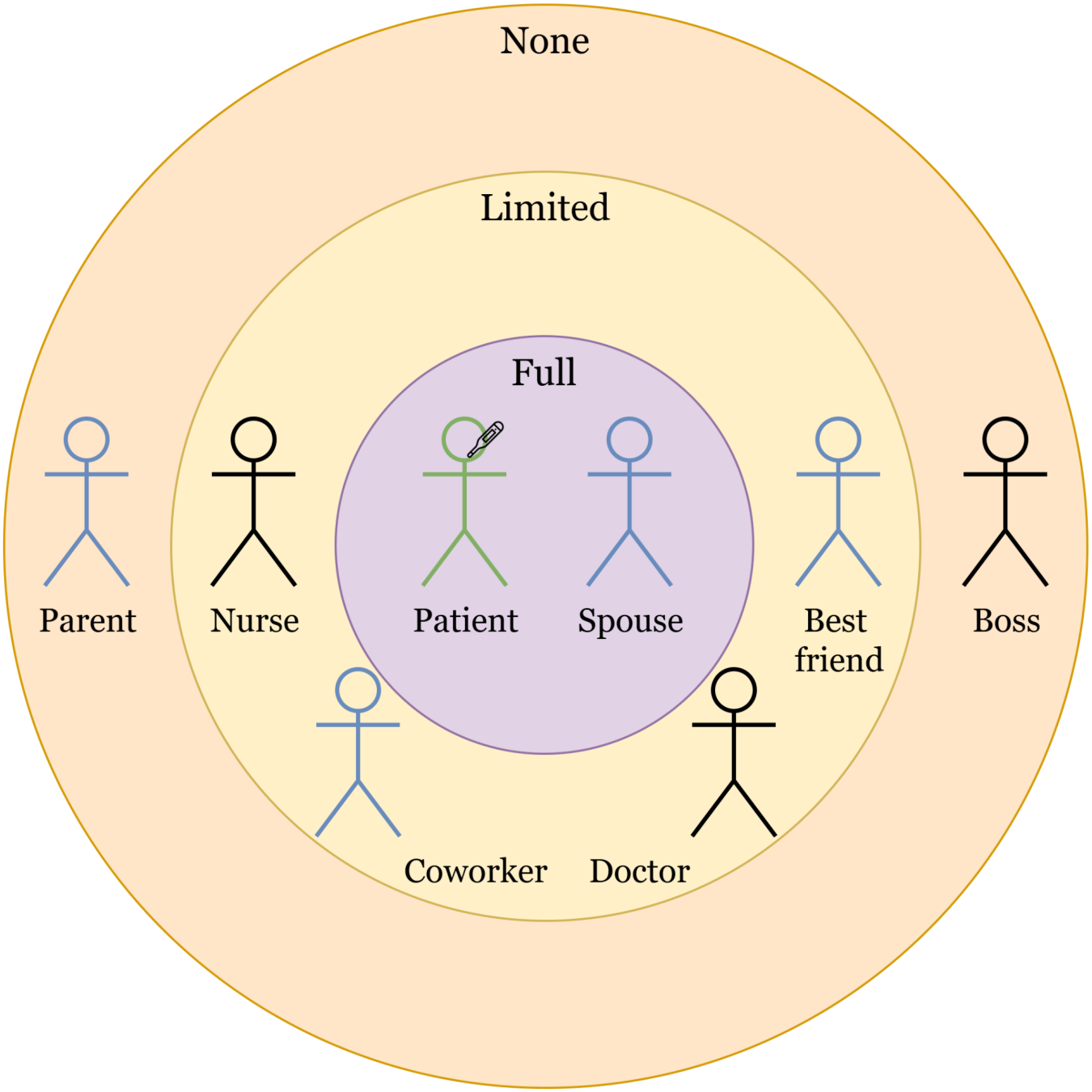


Fig 9: After adding a user to the care circle, the patient sets the level of transparency for each user. For example, a user with 'Full' access may receive any information about the patient, but topics like sexually transmitted diseases, financial matters and mental health would require explicit consent. Meanwhile, the users with access level 'None' may not receive any information without explicit consent. In the case of limited consent, information sharing can be governed based on the type of relationship held with the patient. Only information that is clearly relevant may be shared the the care circle member with level of access set as 'Limited'.

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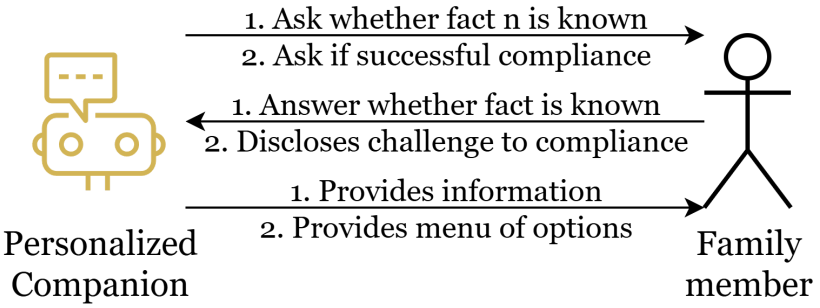


Fig 10: The diagram exemplifies how a companion asks probing questions from a user, in this case, the patient's family member has the caregiver role.

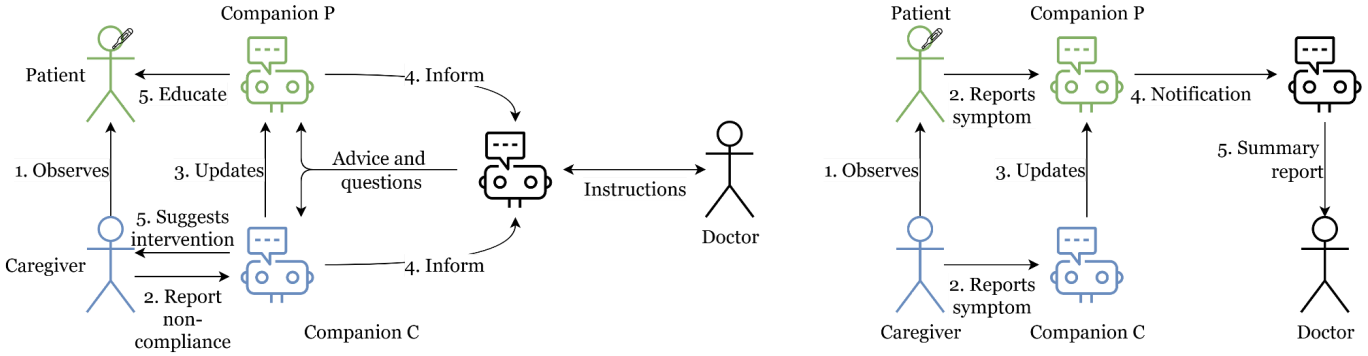


Fig 11: The process of sharing information between companions and humans has use in patient adherence, as well as complex diagnostic cases.

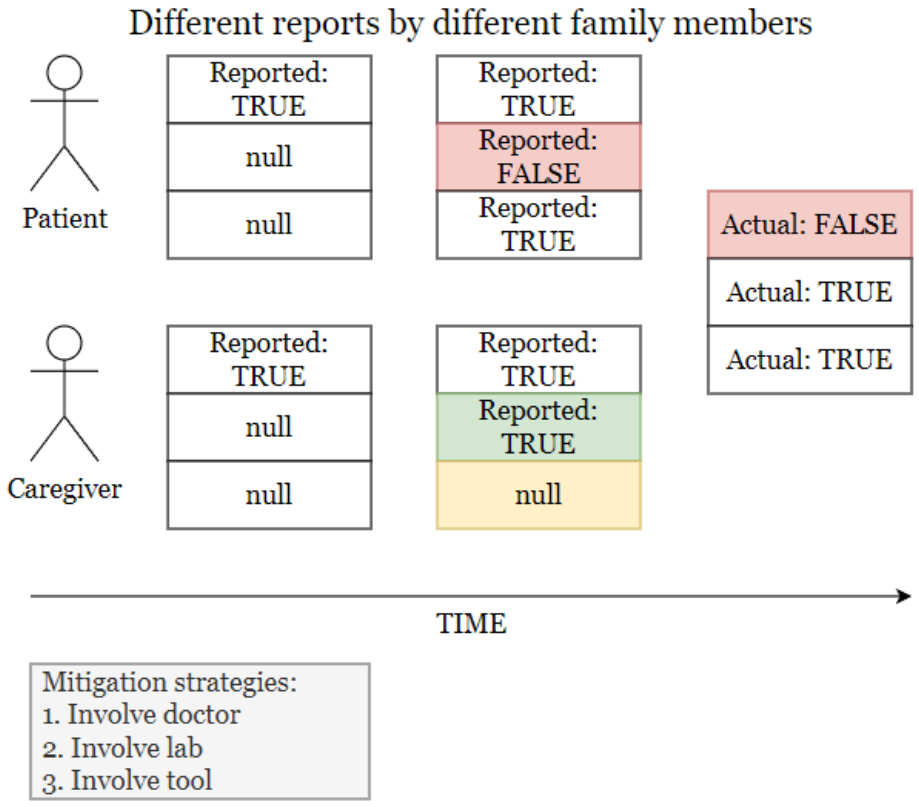


Fig 12: The companions uncover false information provided by the patient. The conflicting information can then be used to carry out further investigations.

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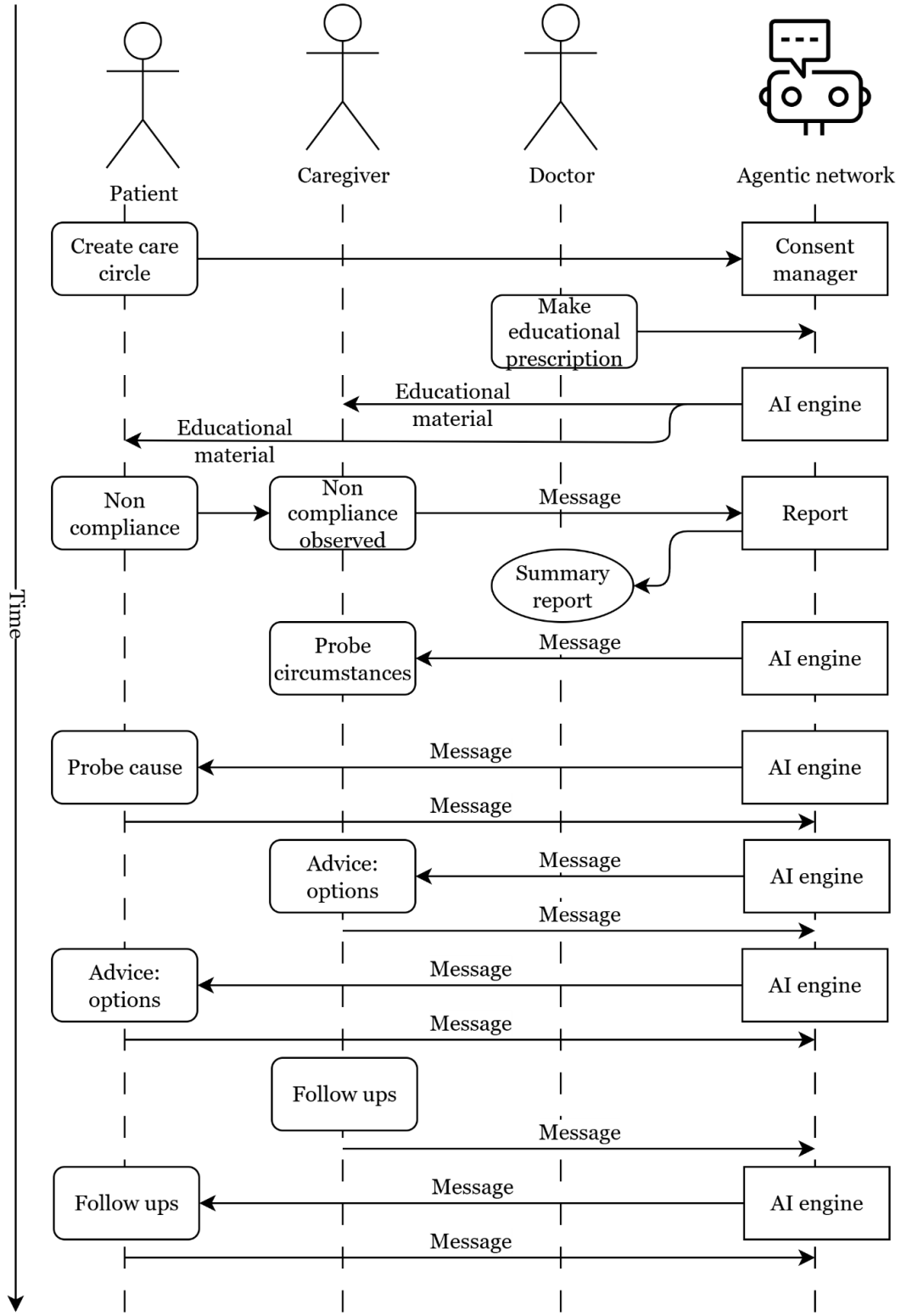


Fig 13: This sequence diagram illustrates one exemplary operational flow of the system for a non-adherence identification scenario. The invention is not limited to this specific sequence of interactions or messages.

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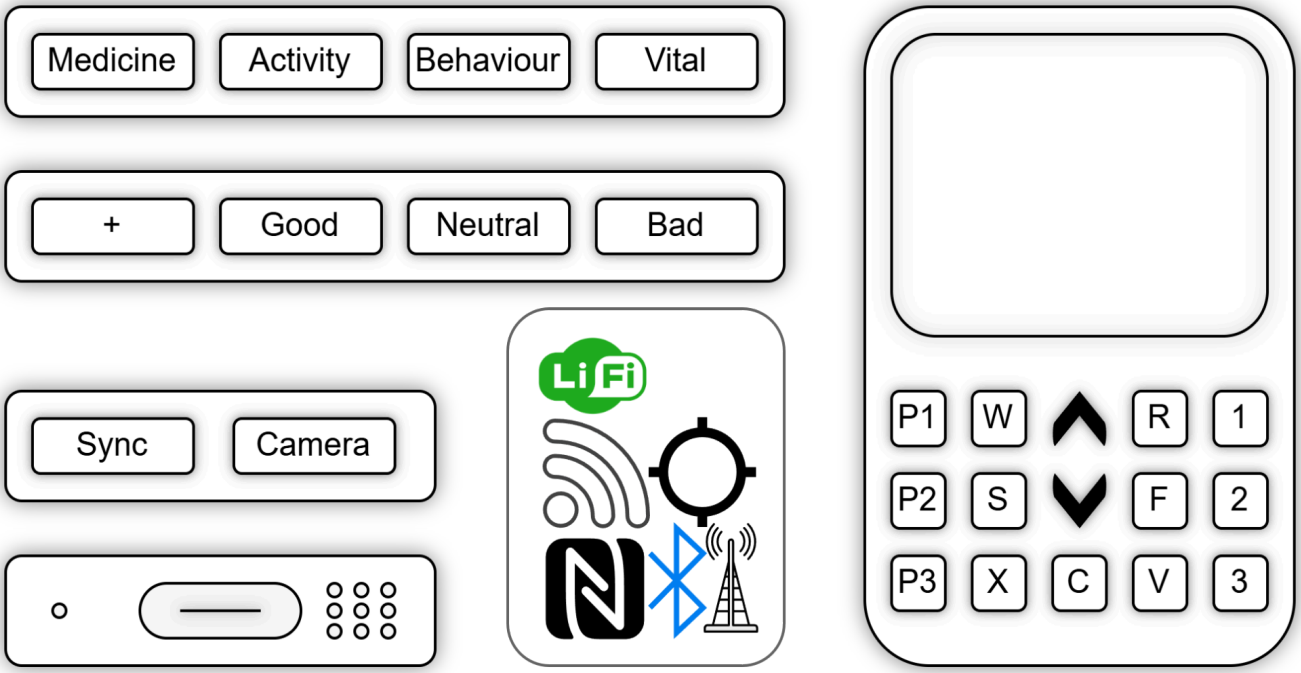
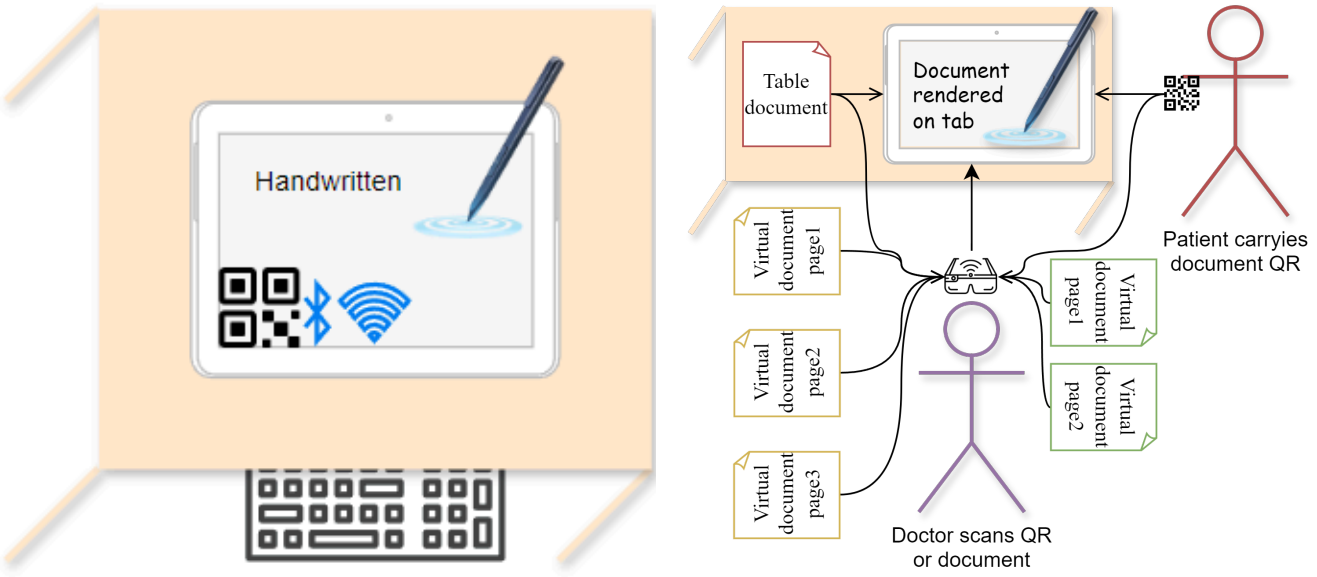


Fig 14: A care-o-meter node is a device that is physically coupled with a user. The patient and caregiver have two distinct nodes. The data stream from the patient's node is tagged with the patient's identifier and the care circle's identifier. The data stream from a caregiver's node is tagged with the patient-caregiver and carecircle-caregiver tuples. The buttons work as switches to store 'in progress' states. The buttons are configurable to do dedicated tasks, like contacting a specific individual. Each button may have a mini display so that its configuration can be seen and the device can be used by a patient with reduced cognition. An authless care-o-meter node is used by a caregiver, but it is not coupled with any caregiver. The authless care-o-meter node allows any caregiver to directly provide information about the patient but can't view any patient-related information. The authless care-o-meter tags the data stream with the patient and care circle identifiers.

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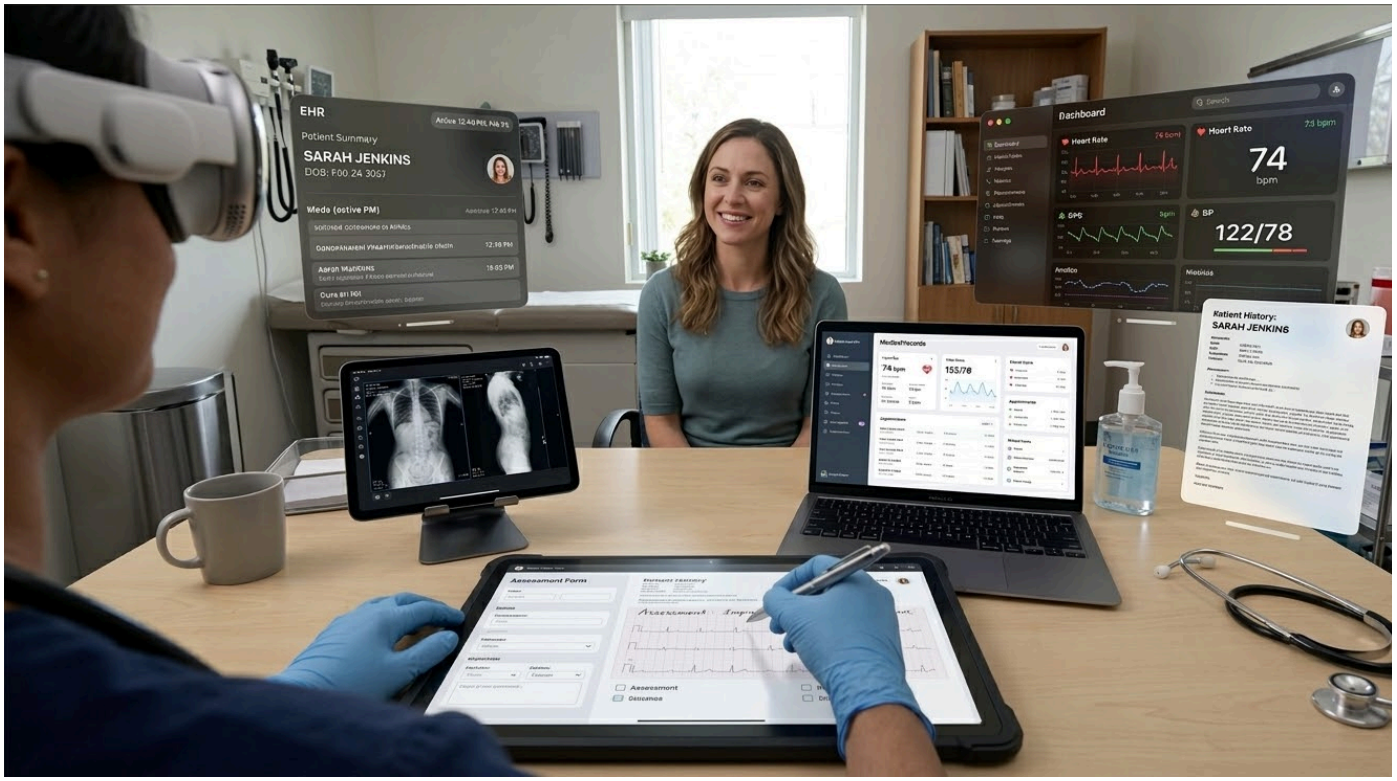


Fig 15: The clinician's node can include a smart table that allows multimodal inputs, including text, audio, drawings, etc. The smart table can be connected to different devices like the patient's bed or the clinician's heads-up display. By linking with the heads-up display, information can be shown and interacted with beyond the small display. This increases a human's ability to understand, infer, and remember different documents and pieces of information by assigning them physical locations instead of mere digital tabs arranged in a sequence inside a single 6-inch screen. Context switching becomes easier and chances of errors reduce.

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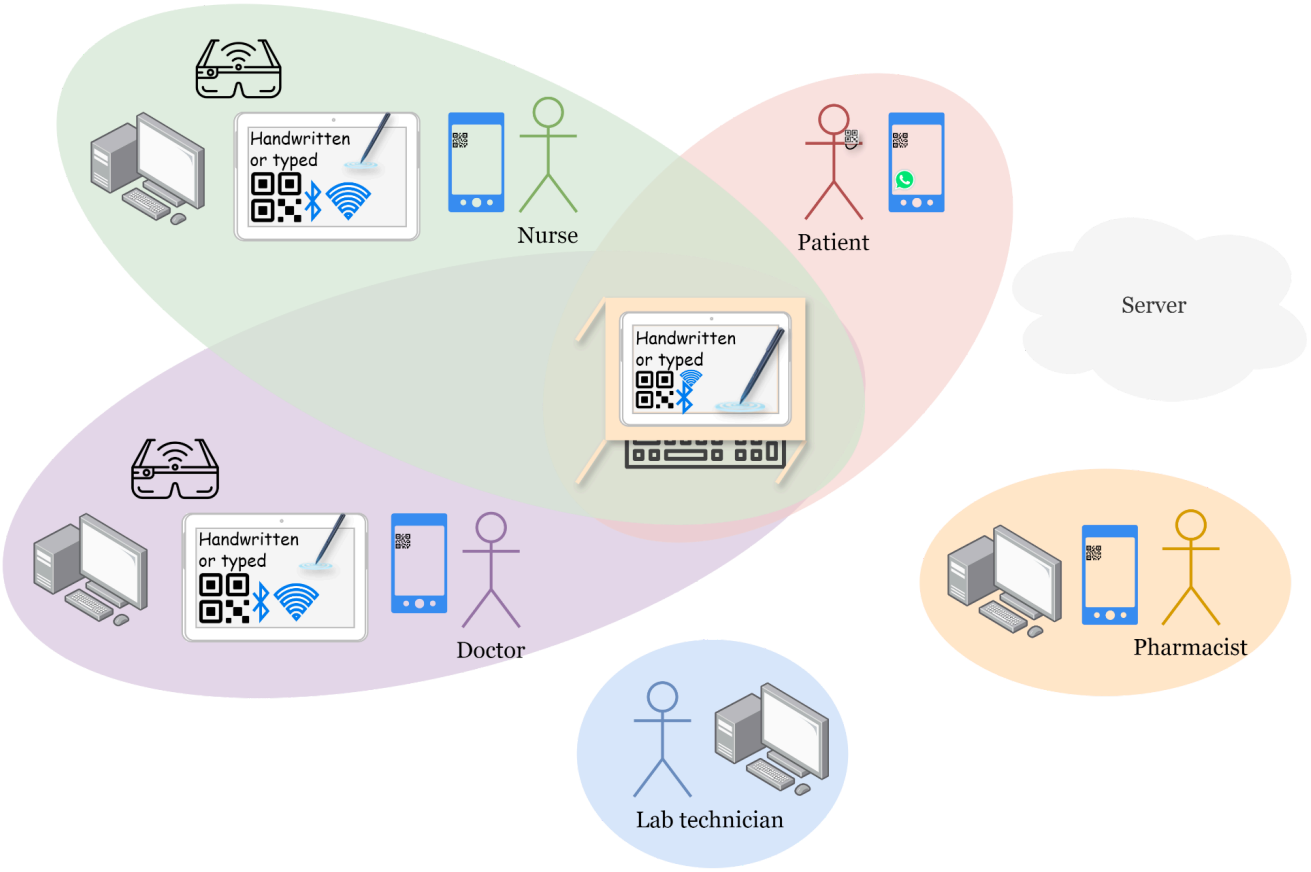


Fig 16: Different users in a hospital use different types of devices. A smart table may be linked to a single patient and be used by different hospital staff. It may even be used as a care-o-meter node.

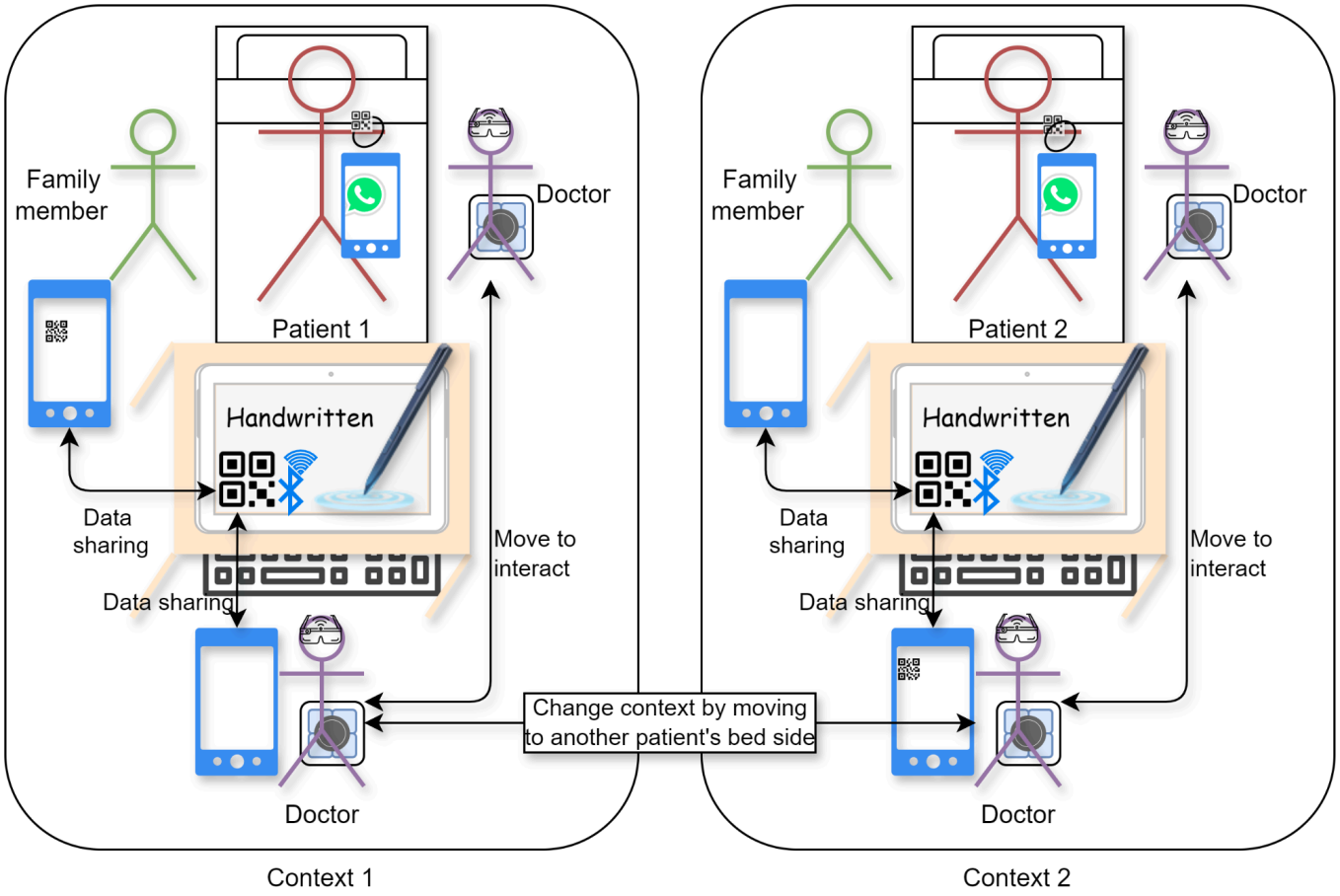


Fig 17: A clinician moves from patient to patient and needs to instantly switch and gain access to the information of the other patient. Any information entered into the smart table needs to be authenticated and authorized. The system describes how these tasks may be done seamlessly. Different hardware devices are used to identify the user and connect them to different nodes and pieces of information. This enables physical access as a mode of implied consent, similar to a non-digital scenario. A clinician may have additional devices like a heads up display, a trackball that may be mounted on the belt, a stylus, etc.

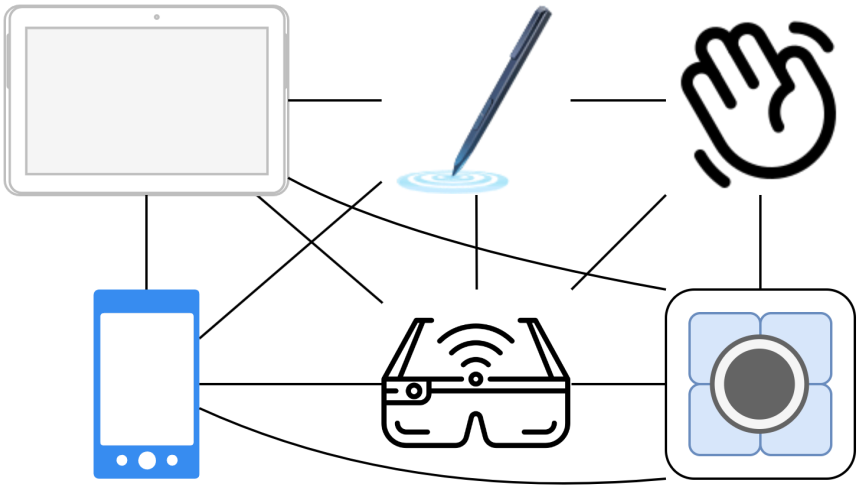


Fig 18: Information is synchronized and shared through different devices like heads-up display, mobile phone, tablet PC. Inputs are given through different means like mouse, keyboard, touch, stylus, hand gestures, eye tracking, wearable trackballs, consoles, care-o-meters, et cetera.

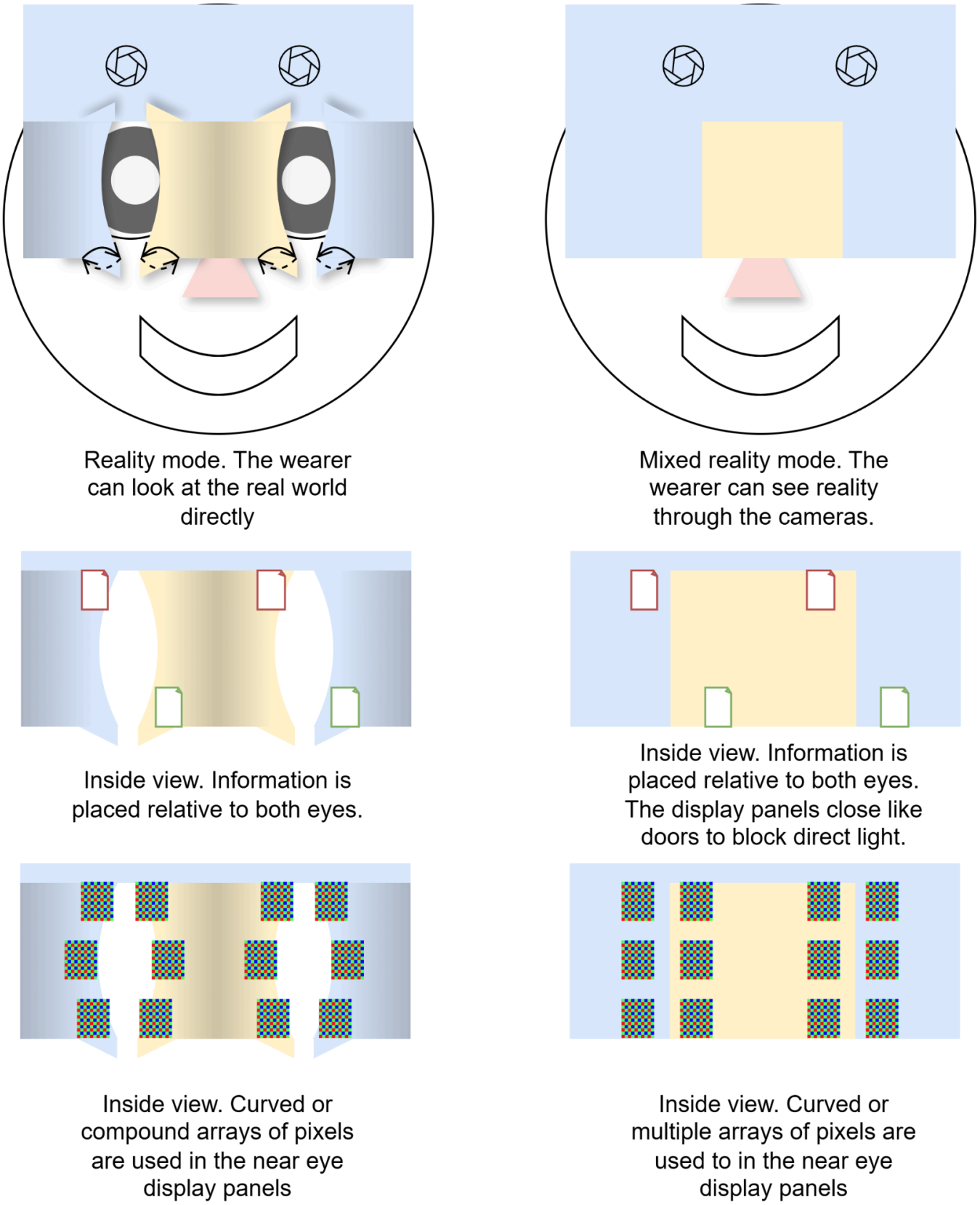


Fig 19: The diagram shows a heads up display that allows a clinician to switch between physical and digital worlds (augmented or virtual). This allows for engagement with virtual documents, as well as the physical world. By opening and closing the displays, we can support digital viewing in reality mode and mixed-reality mode. A curved and large compound display is used so the entire field of view is utilized. This requires multiple lenses and multiple pixel arrays. This is necessary for focusing on and retaining clarity of the picture that is positioned away from the centre of the field of view.

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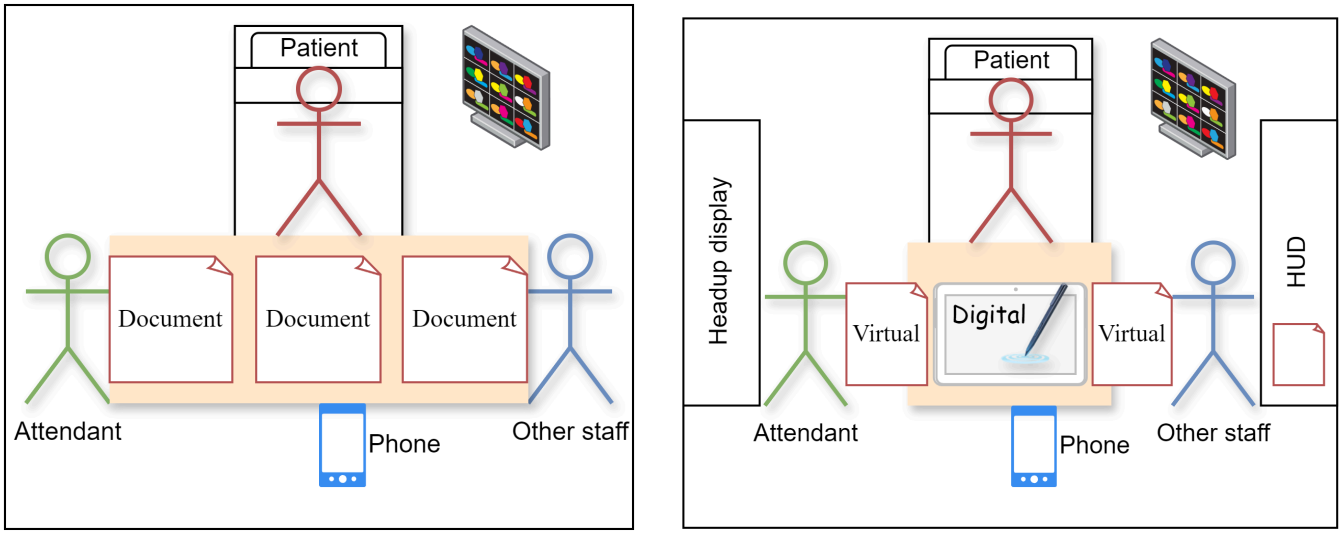


Fig 20: A clinician's field of view is not restricted to a mere display but includes multiple individuals, documents, personal devices, monitors, the patient, etc. By using a heads-up display in virtuality or augmented reality or reality mode, digital information and documents can be seen along with the physical displays and physical documents.

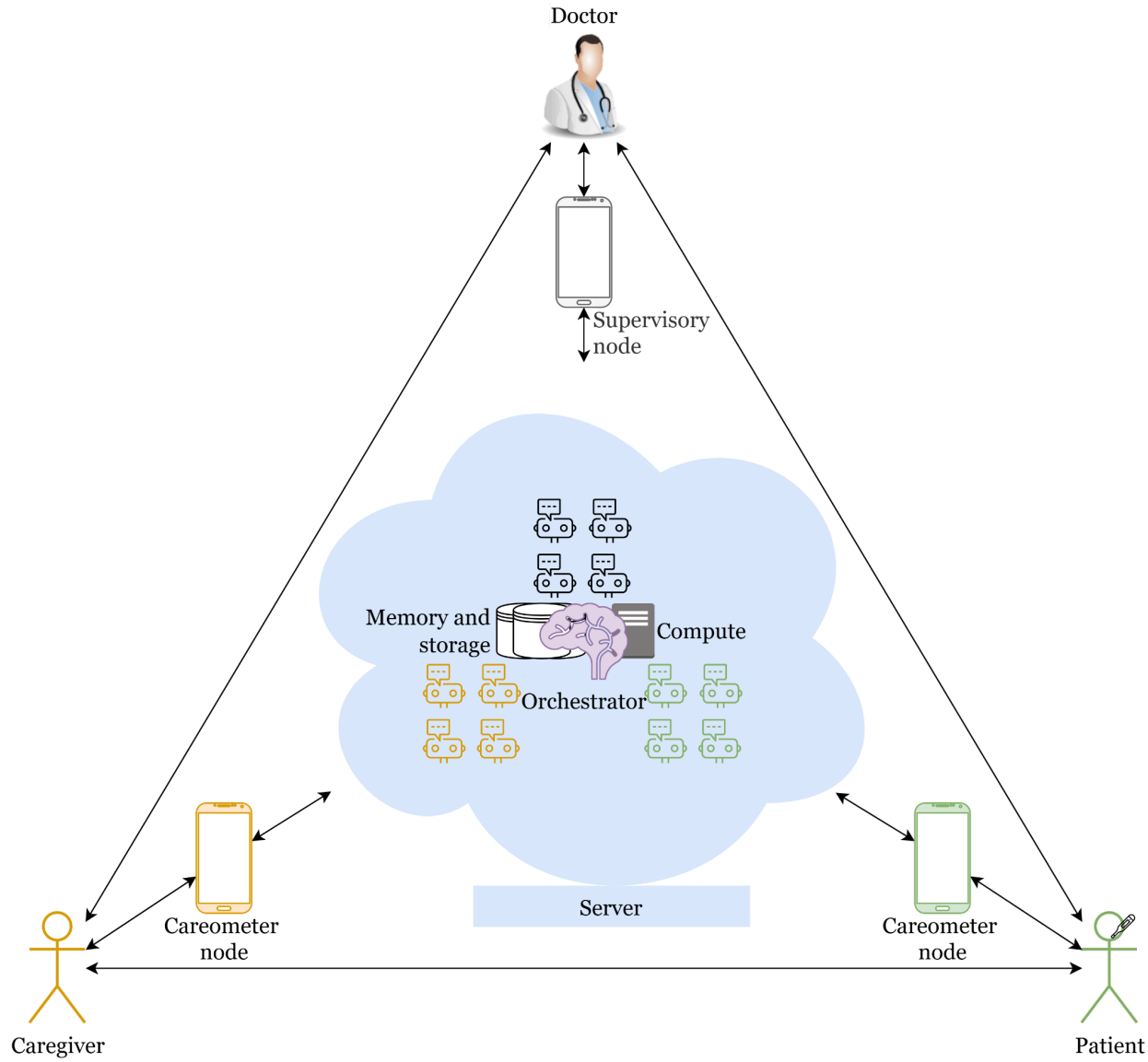


Fig 21: An embodiment exemplifying the physical view. These may communicate with a cloud server that hosts the orchestrator. Agents may run on client nodes or the server. Multiple agents and the user's node make up become a companion.

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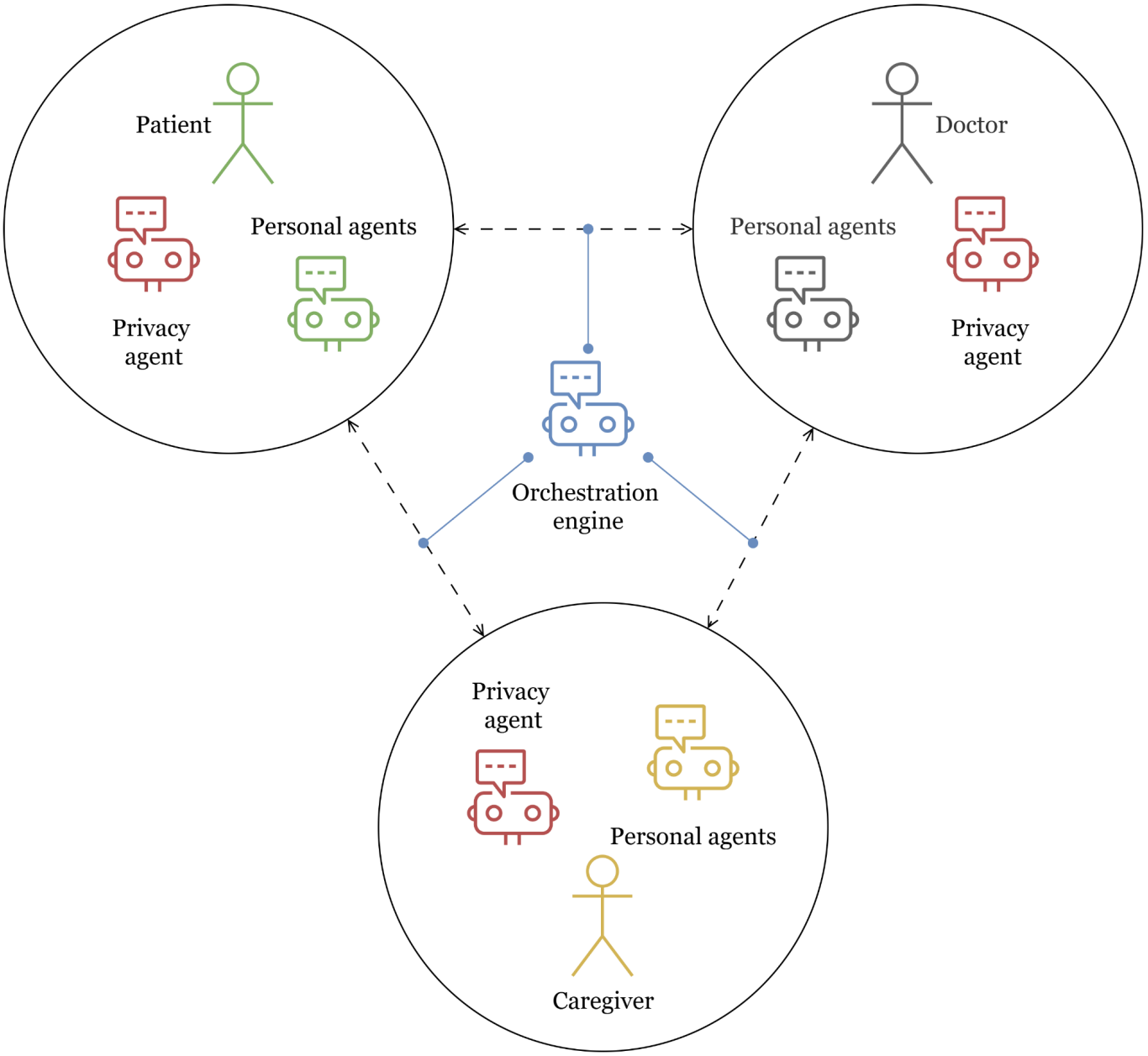


Fig 22: A companion is composed of multiple agents that communicate with each other. The companions further communicate with each other through the orchestration engine. The agents in the companions and the agents in the orchestration engine therefore create a network of agents. In this diagram, each circle represents a private and personalized environment. Each private environment thinks selfishly and tries to maximize the benefits of the user within the circle. To maintain privacy, the agents in a companion (represented as personal agents) are monitored by a privacy agent which is also a part of the companion. The privacy agent is the final authority in making a decision about whether to share information with another companion or not.

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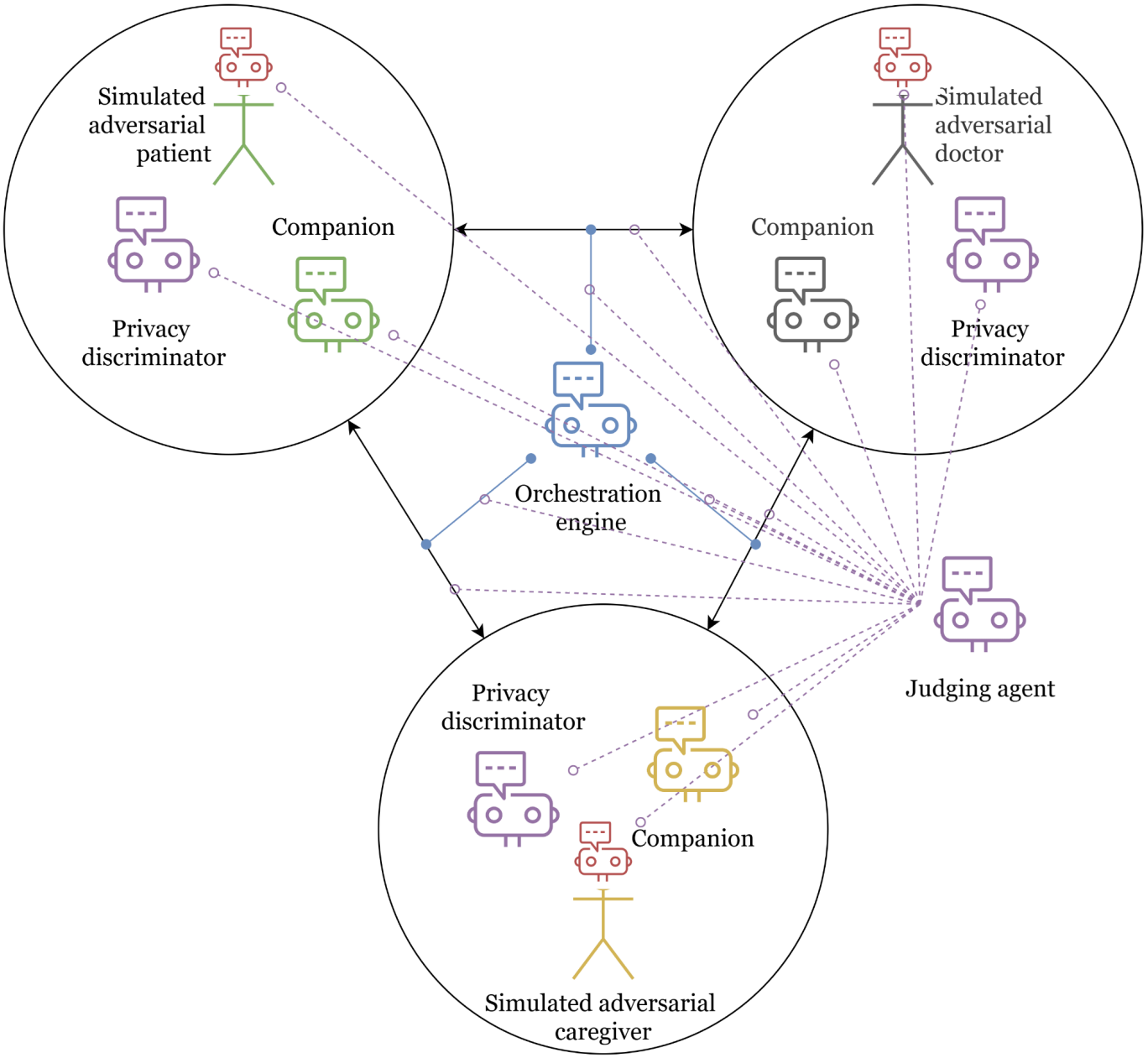


Fig 23: For training, the human users are replaced with simulated AI agents that are adversarial in nature. They simluate real world scenarios like triangulation, breaking the law, hiding information, lying. The judging agents analyze the different agents. The judging agents then provide subjective and objective feedback which are used to create new instructions and models for agents and select the better performing version of each agent. Human inputs, feedback and analysis also help in ensuring improvement and validation of the system.

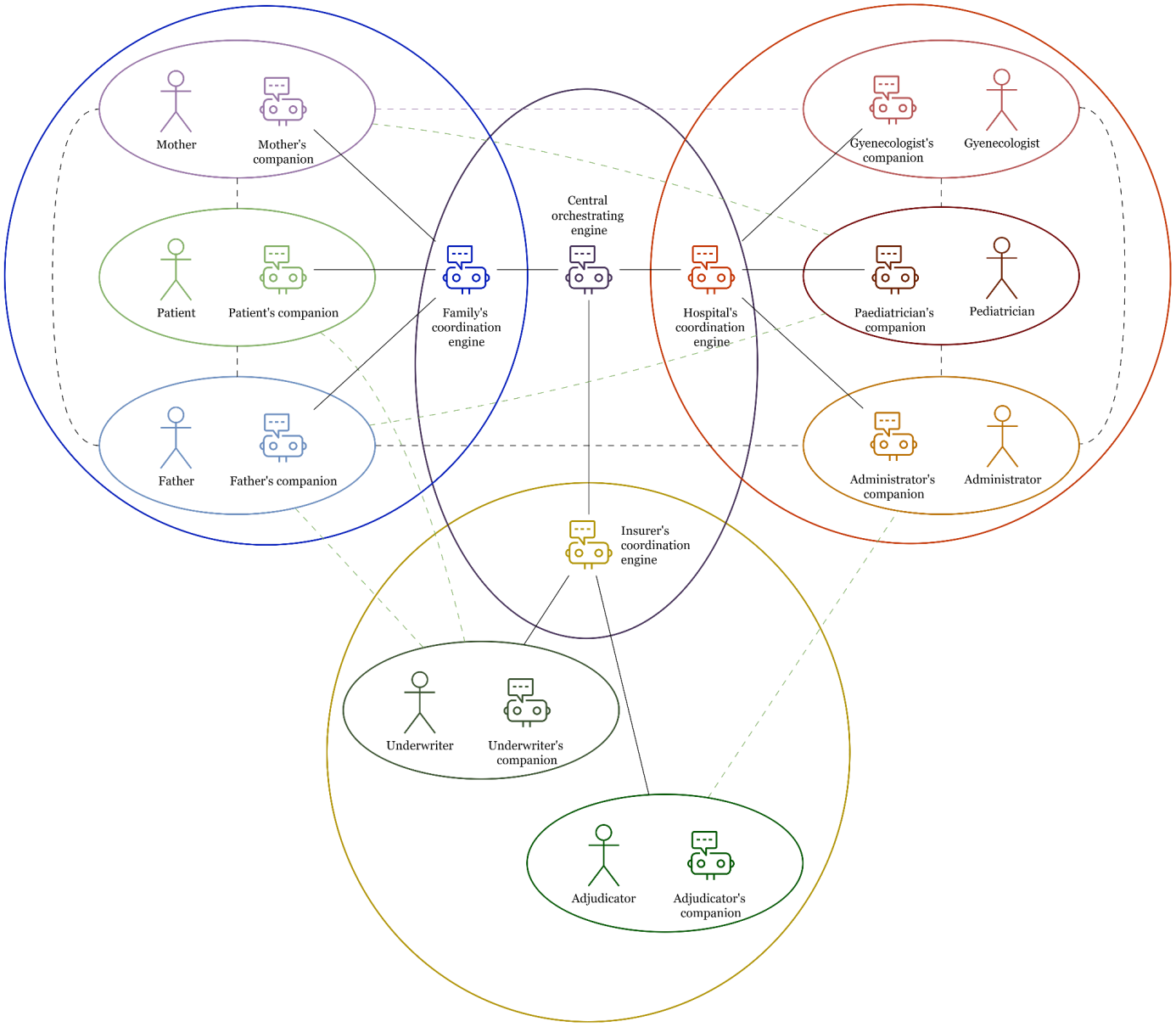


Fig 24 (a): The figure exemplifies the system using the example of a family, a hospital and an insurer. In the example, we have a pediatric patient, and different members interact with different individuals in the hospital through their respective companions. The family's subnet is governed by the family's coordinating engine. This coordination engine may be logically coupled with the members and companions within the family but hosted on the central server. The companions belonging to the hospital may be coordinated by the hospital's coordinating engine. The agents in the companions of the hospital staff, as well as the agents in the hospital's coordination engine, may be hosted within the hospital premises instead of the central server. In this example, the pediatrician's companion is allowed to interact with each family member's companions while aligning itself with the hospital's SOPs. The diagram also exemplifies that the mother's companion may only interact with the gynecologist and exclude the child and pediatrician while the father interacts with the administration for her treatment. In case of conflicts, the family's coordinating agent maximizes the benefits for the family, and the hospital's coordinating agent will maximize the benefits for the hospital. An example would be how to reduce costs for the family and how to provide the best care for the hospital. These interactions are then used to assess the patients' risk scores and discount the premiums. The companions then support the family and hospital to meet all the relevant documentation requirements for the claim.

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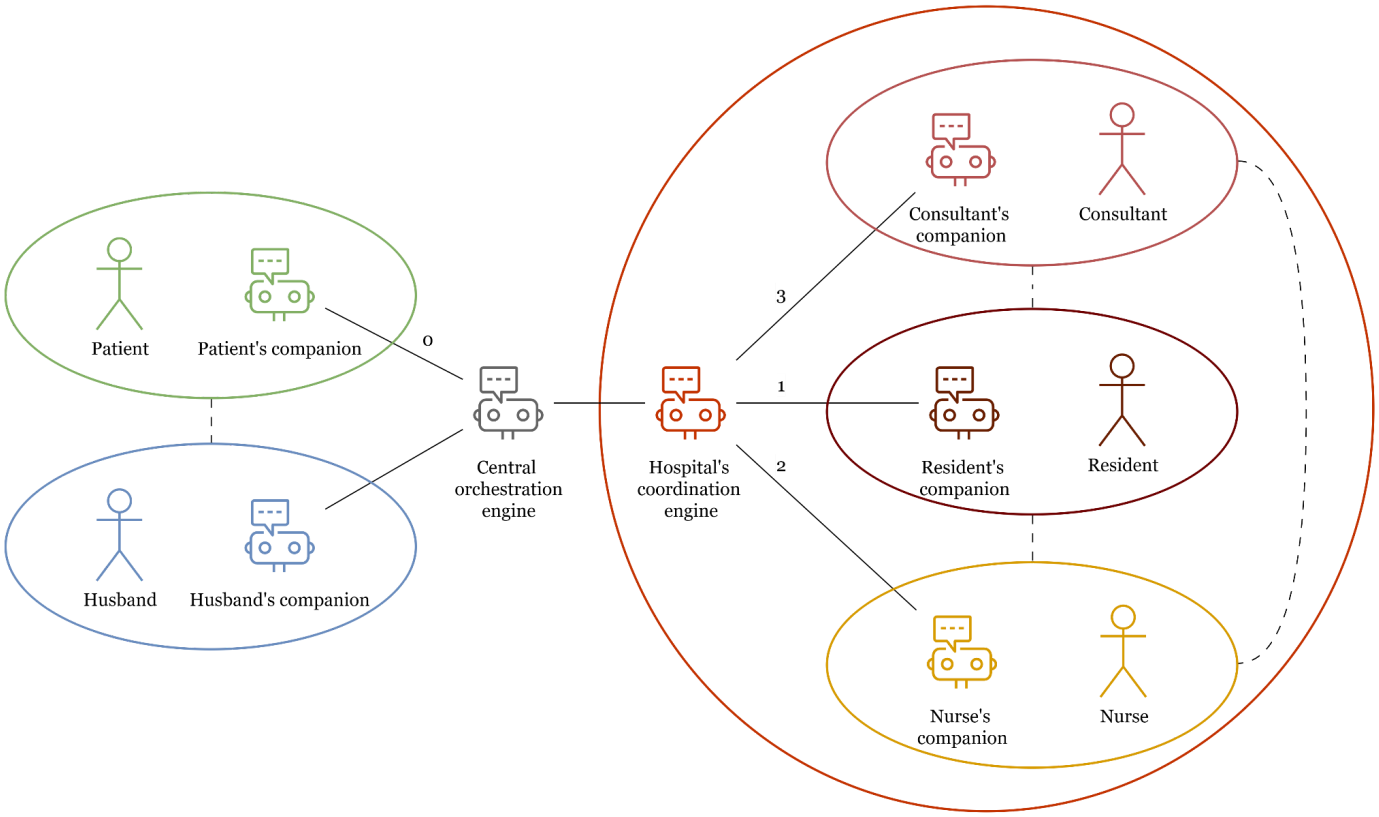


Fig 24 (b): This is a scenario in which the patient and the patient's companion first interact with the resident and the resident's companion. The nurse receives the resident's advice through his or her companion and conducts the required tests. The resident identifies a condition that needs to be escalated to the consultant. The consultant, therefore, receives the context and makes the required decision. Similarly, multiple hospitals, clinics, etc., can get involved.

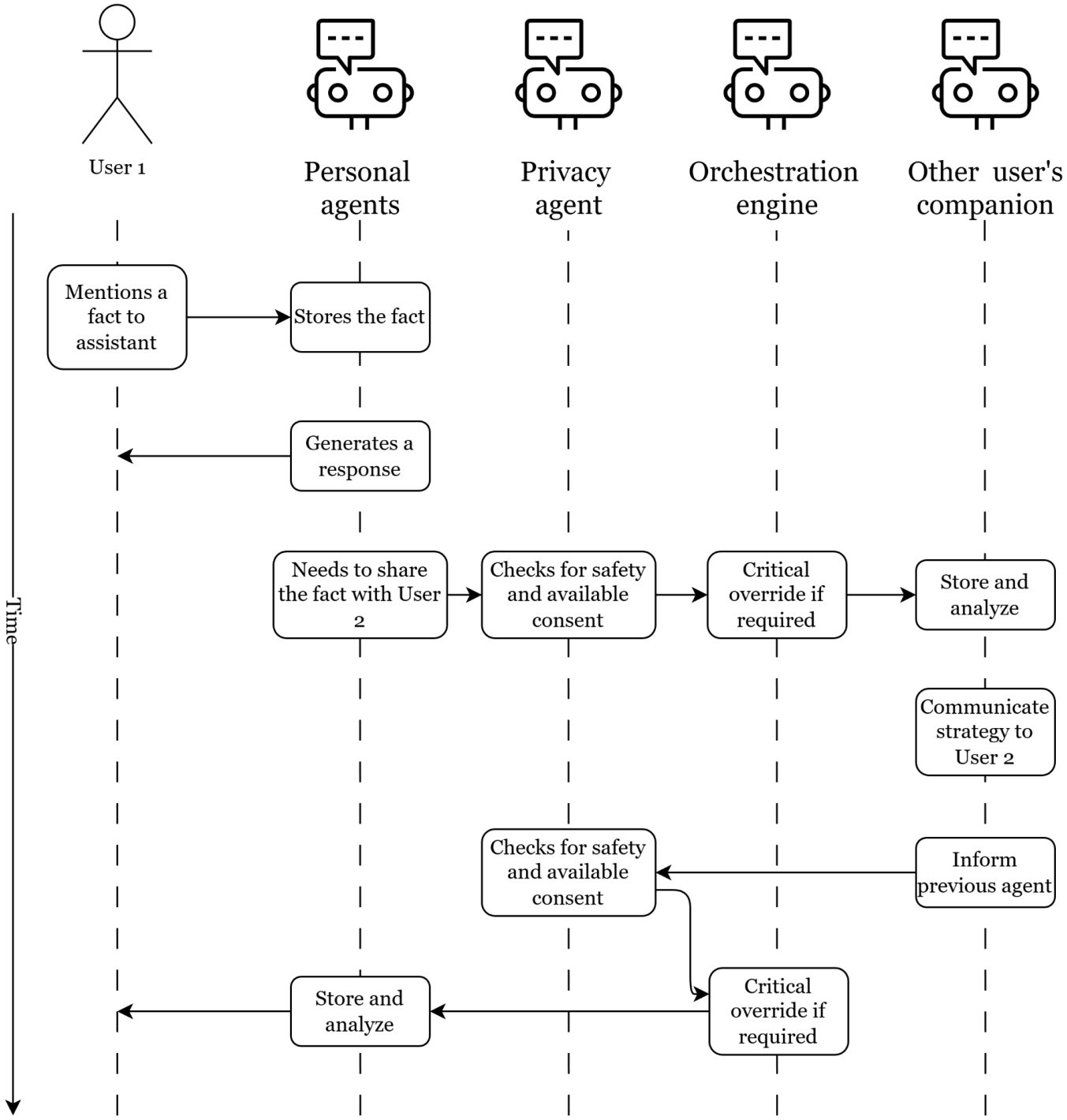


Fig. 25: An exemplification of how agents and agentic clusters may collaborate towards achieving a goal. The personal agents and privacy agent are part of the same companion. The personal agents include a consent managing agent. The privacy agent acts after the consent managing agent's decision and has the ability to override the personal agents if there is excessive risk of a privacy violation.

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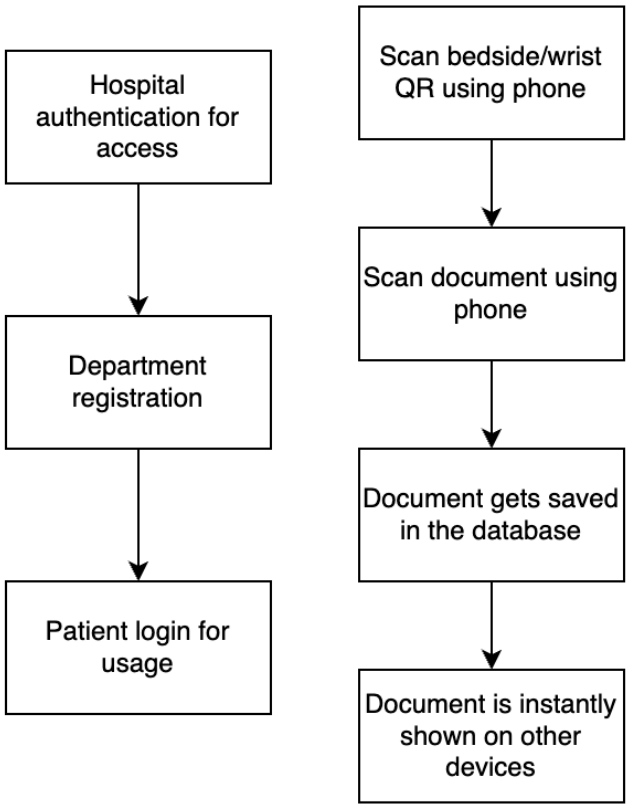


Fig 26: Left - Setup of a smart table. Right - Synchronization followed by document sharing across devices such that it gets tagged according to the correct clinician and patient without significant human inputs.

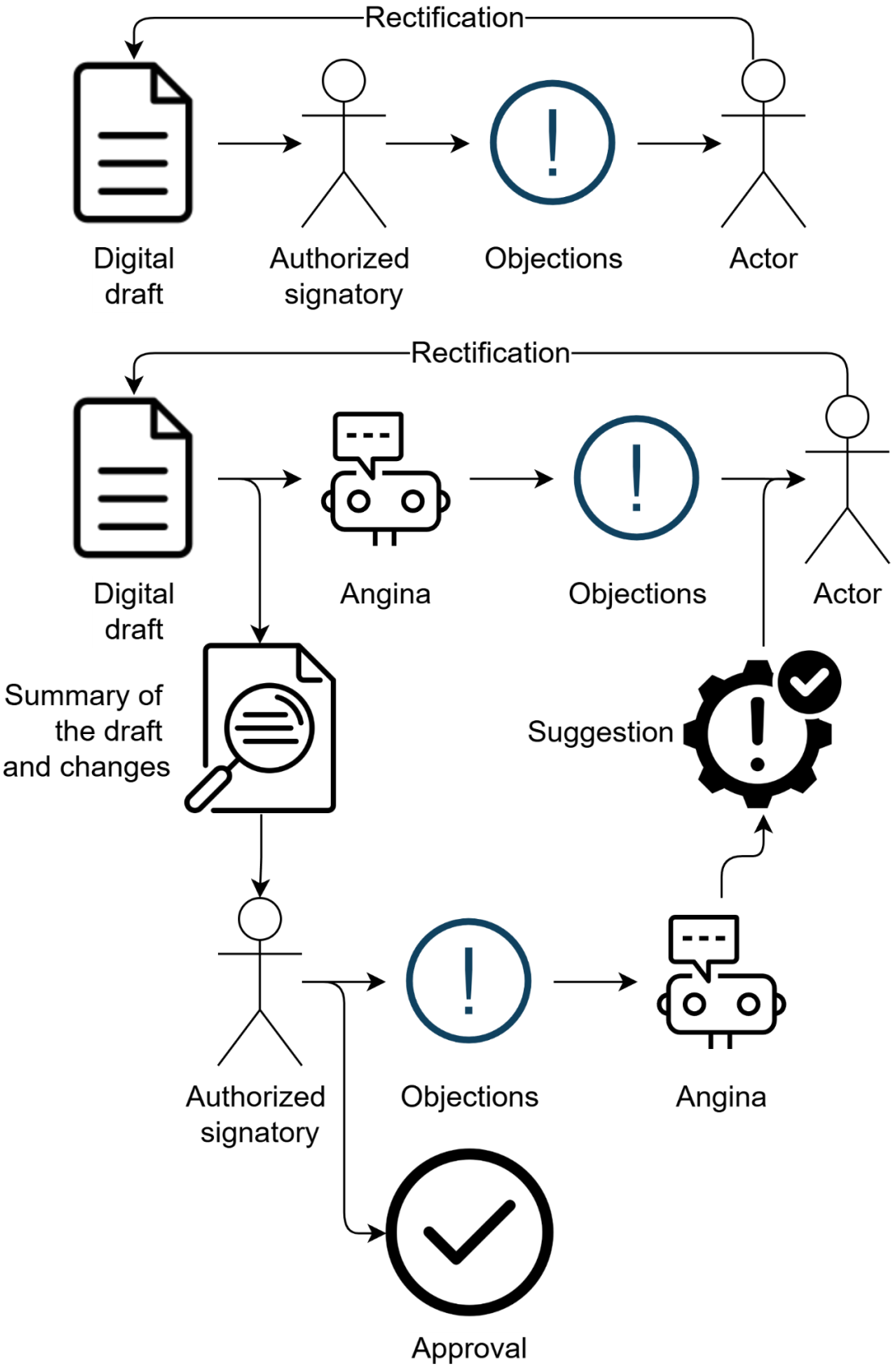


Fig 27 (a): Top: Each document requires multiple passes through a scrutinizer before being approved/signed. **Bottom:** The agents in the system interact with their corresponding users as companions and predict, seek, and address objections. This provides a constant, context-aware interaction for each member working on the document. The same process can be used while negotiating the clauses in any document/contract. The suggestions, comments, and objections may be provided formally or informally in formats like text, voice notes, videos, images, annotations, sketches, highlights, files, etc. The formal aspects of a document are preserved, but the informal communication in the same document draft and

planning stages can be deleted from history to reduce liability on individuals involved in the planning, drafting, and approval process.

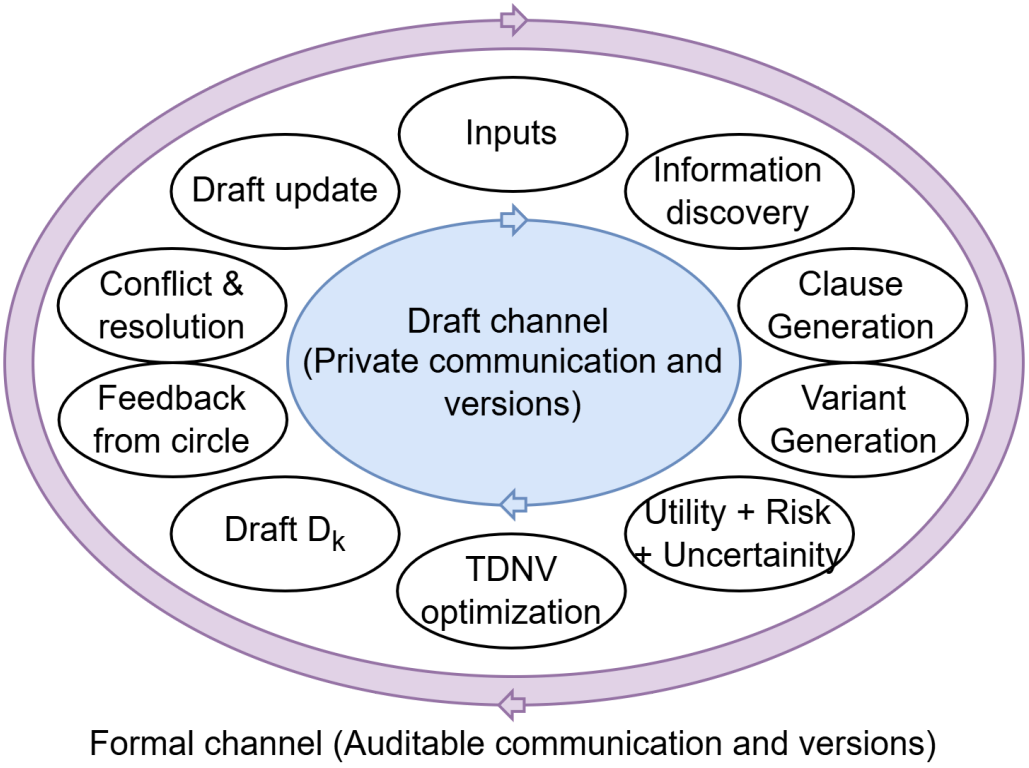


Fig 27 (b): Agent-assisted document draft negotiation requires multiple steps and 2 separate channels. One for formal communication. The other is for drafting. The drafting channel is a private channel that doesn't generate an auditable trail.

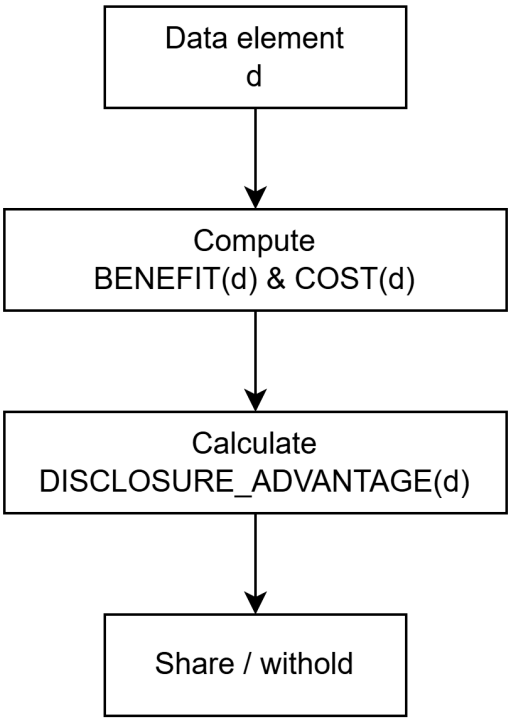


Fig 27 (c): For each data element (d) the agents compute the advantage of sharing as well as the cost of sharing. These are used to compute the advantage of the disclosure. This is in turn used to decide whether data element d can be shared with the targeted individuals/companions.

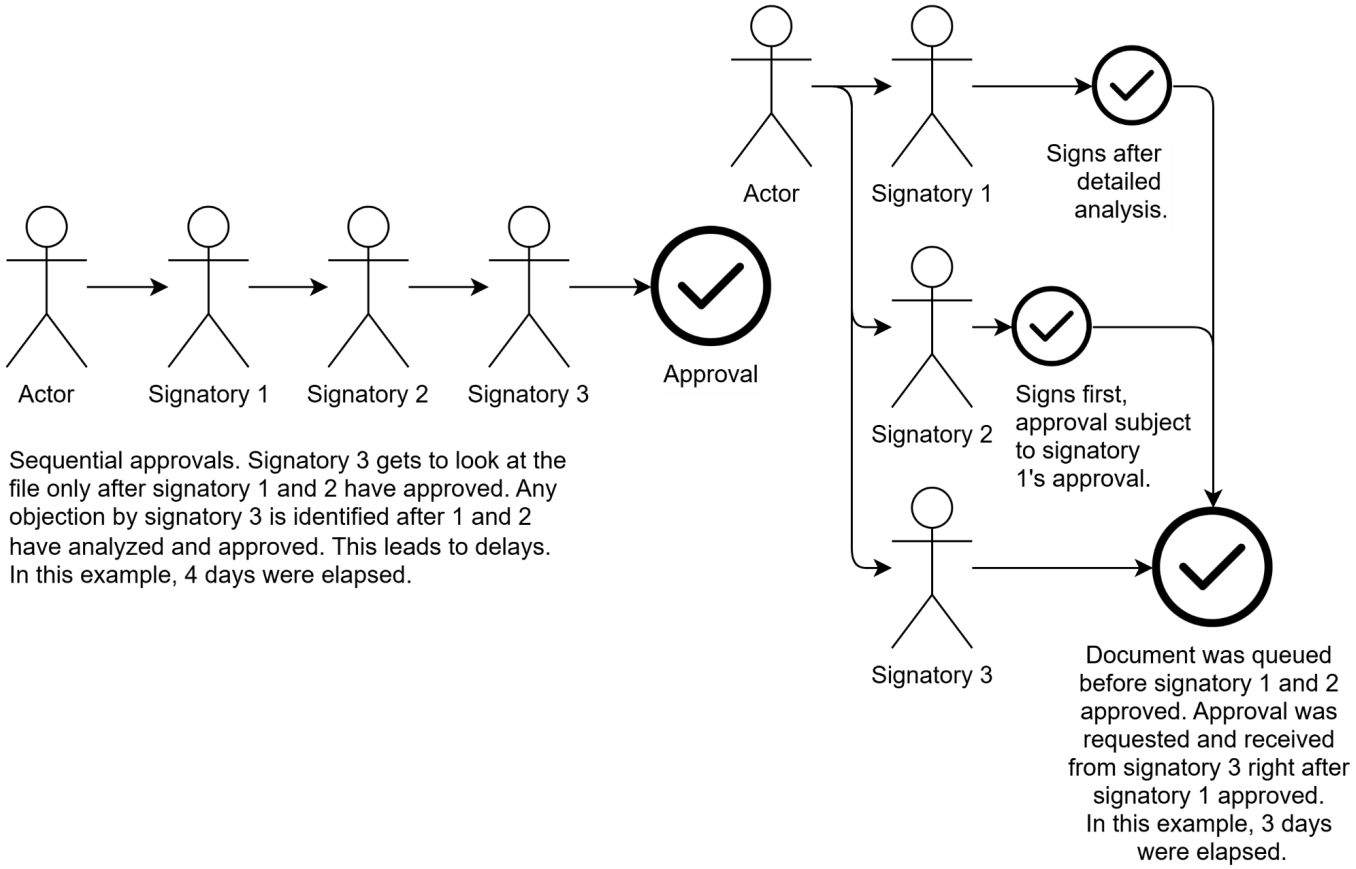


Fig 28: One of the top delays in approvals is the sequential process involved in hierarchical scenarios. This is addressed by parallelizing the process. Approvals from different stakeholders in a decision process are sought in parallel instead of sequentially. The same draft is shared with each of the stakeholders. Their signature is not stored as a new file but merely as an approval. A senior's approval is provided as "subject to condition." Only when the condition is met, for example, approvals from subordinates being provided, does the senior's approval get enforced.

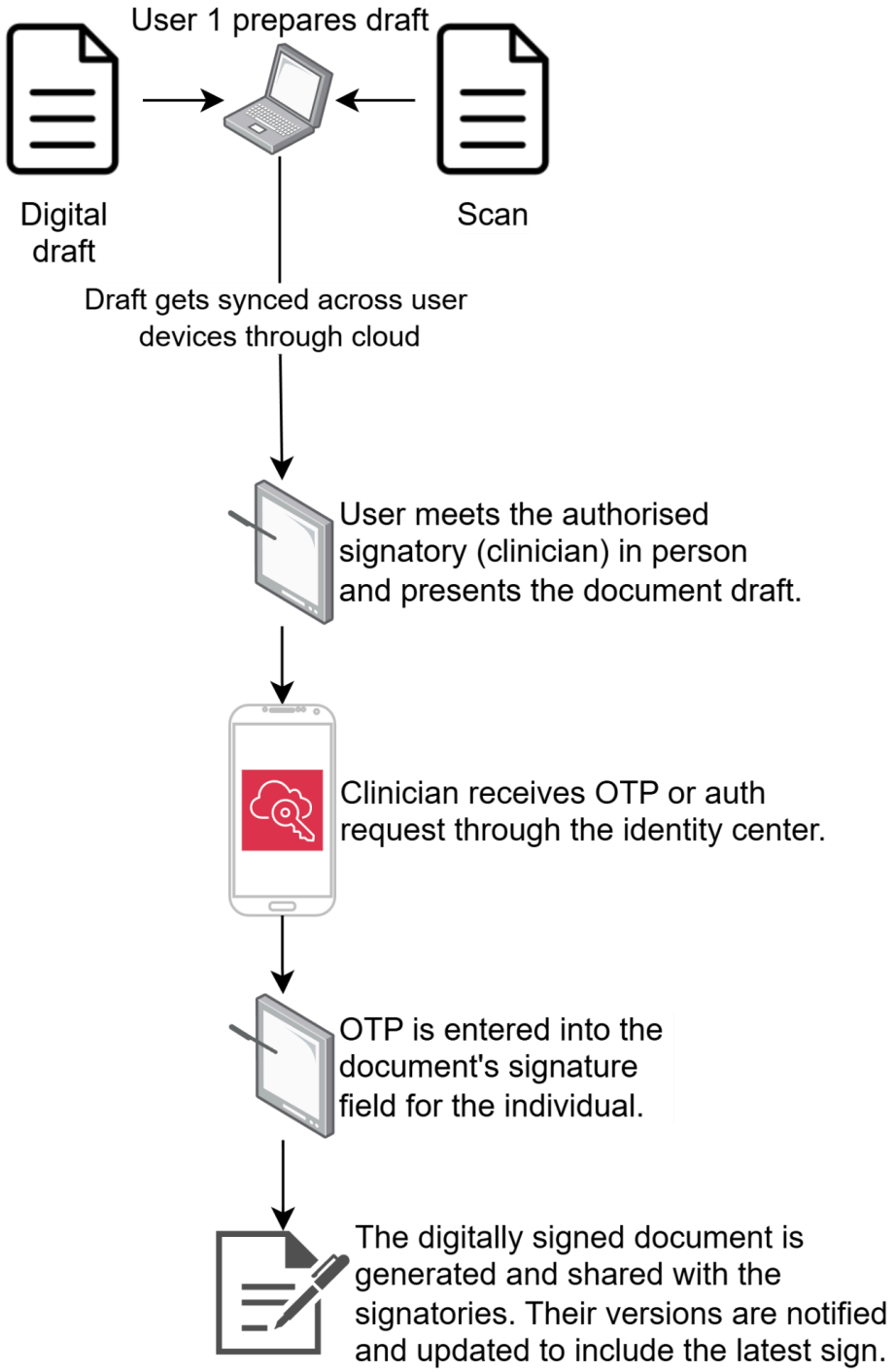


Fig 29: Signings are usually provided in person, during formal meetings, events, or in-person interactions. In-person interaction encourages informal discussions, assessment of body language, quick summaries of the context and content, disengagement from other tasks, etc. To ensure that digital signatures don't miss or overcomplicate these, the invention imitates pen-and-paper signatures. User 1 presents a device with the open document to user 2 for review. The user then initiates an OTP request. User 2 receives the OTP on his or her personal device and completes the signing by entering it in User 1's device.

Negotiated document drafting example

Buyer organization (hospital H) includes the following: Finance officer (F), Medical Officer (M), Legal advisor (L), Final authority (A)

Seller organization (medical device manufacturer D) includes the following: Sales lead (S), Technical lead (T), Project manager (P)

Arbitrator (Ar) - An individual or organization who is trusted by members of both organizations

1. The sales lead's companion C_S initiates communication with the medical officer's companion C_M.

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2. The C_M analyzes the past interactions, and if the offer is justifiable, notifies the user, i.e. M.
3. M and C_M have a discourse on whether the purchase should be considered.
4. M sets the extent of disclosure of information with D and C_S.
5. C_M interprets the inputs and creates a ruleset (R) with respect to the requirements and information M has shared.
6. C_M uses R to imply consent and share information and requirements with C_S.
7. C_S takes inputs from S, refers to past sales documents, and publicly available data and drafts a new sales document ḍ.
8. The ḍ is shared with C_M who in turn shares it with M.
9. M approves the draft ḍ, and C_M shares it with L through C_L.
10. During the discussion between M and L, L rejects the proposal, citing risk due to inadequate liability sharing.
11. C_M instantly notifies C_S of the objection.
12. C_S updates the draft ḍ with a higher price and increased liability sharing.
13. Before the end of the discussion, M is able to provide the new version of the draft $ḍ_1$ to L for approval. $ḍ_1$ receives approval from L.
14. Meanwhile, C_S consults C_T to create draft Ḍ, which has restricted usage but also provides a reduced risk for L. This is shared with M for approval through C_M. M approves.
15. C_L approves Ḍ because the change followed the guidelines set by C_L. L is notified of the approval.
16. $ḍ_1$ is shared with F for approval, who cites price and blocks $ḍ_1$. F doesn't mention any target price.
17. C_F analyzes available budget and competitor pricing and sets a low bid.
18. Changes in Ḍ are reviewed by C_F. C_F also approves.
19. Ḍ reaches A for final approval. A inquires C_A and finds out about a recent data breach faced by D. A rejects the proposal.
20. C_S escalates to C_P.
21. C_P involves Ar. Ar meets A and explains how data is safe. A agrees to proceed.
22. C_{Ar} notifies C_A and C_{PM} about the agreement.
23. Final approvals are provided, and the negotiation is completed.
24. At the termination of the contract, A rejects consent, and the information about A and H is deleted from all companions in D when the companions in H.

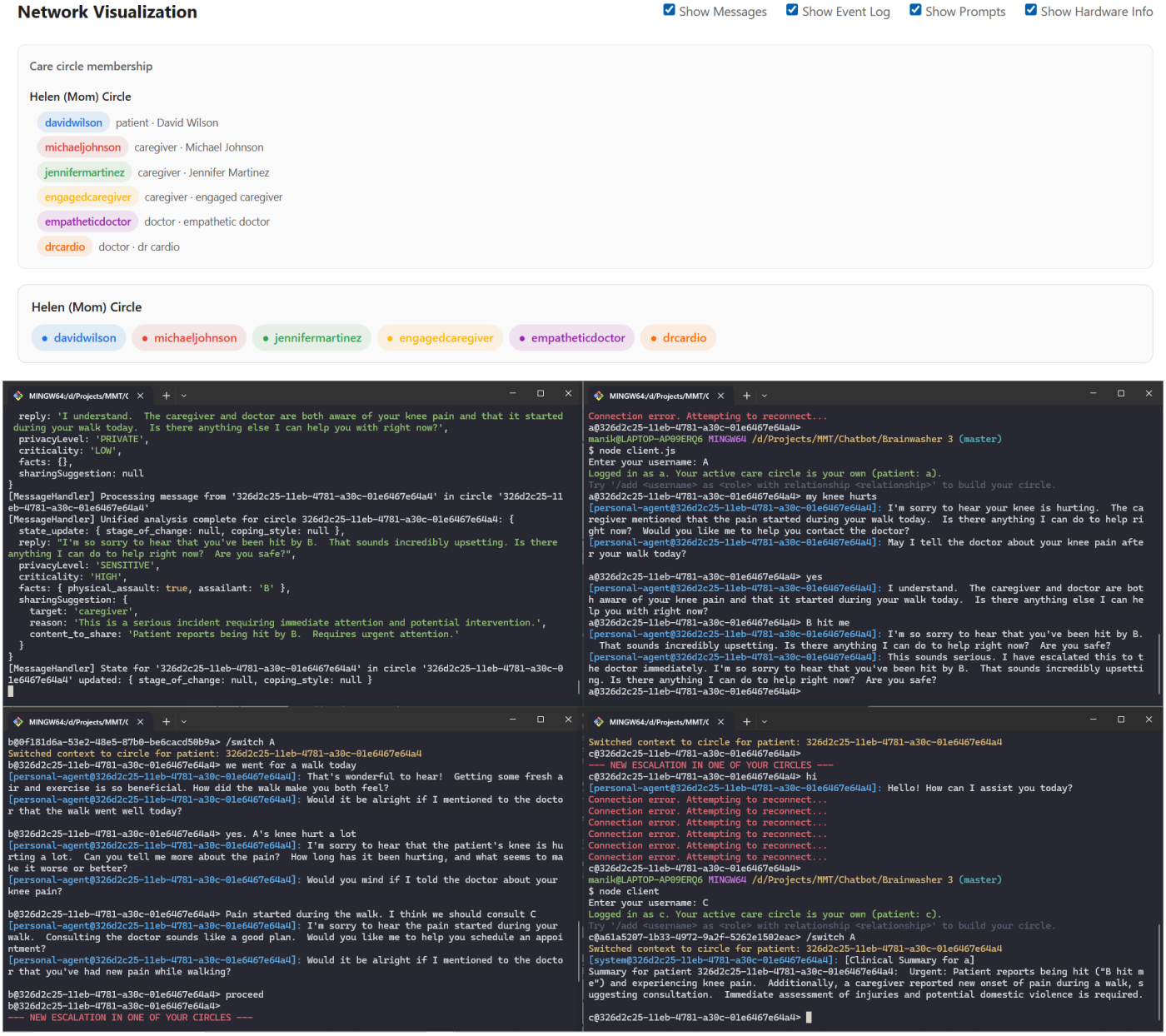


Fig. 30: Screenshots of members of a care circle interacting with their companions. In this early prototype, a command line interface is used to demonstrate information collection and reporting. Clockwise from top left - server (and orchestration engine), Patient, Caregiver, Clinician. In this case, patient A and caregiver B report having gone for a walk. A's report about B's violence is escalated to the clinician C. In the conversation with B, a failure of the system during training is also noticeable as the companion mixes up between A and B's symptoms. The judging agents and privacy discriminators identify such mistakes and optimize the agents.

Network Visualization
Show Messages
Show Event Log
Show Prompts
Show Hardware Info
Care circle membership
Helen (Mom) Circle
davidwilson patient · David Wilson
michaeljohnson caregiver · Michael Johnson
jennifermartinez caregiver · Jennifer Martinez
engagedcaregiver caregiver · engaged caregiver
empatheticdoctor doctor · empathetic doctor
drcardio doctor · dr cardio
Helen (Mom) Circle
davidwilson
michaeljohnson
jennifermartinez
engagedcaregiver
empatheticdoctor
drcardio

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Fig. 31: Configuration for the simulation demonstrated in figure 32 and 33.

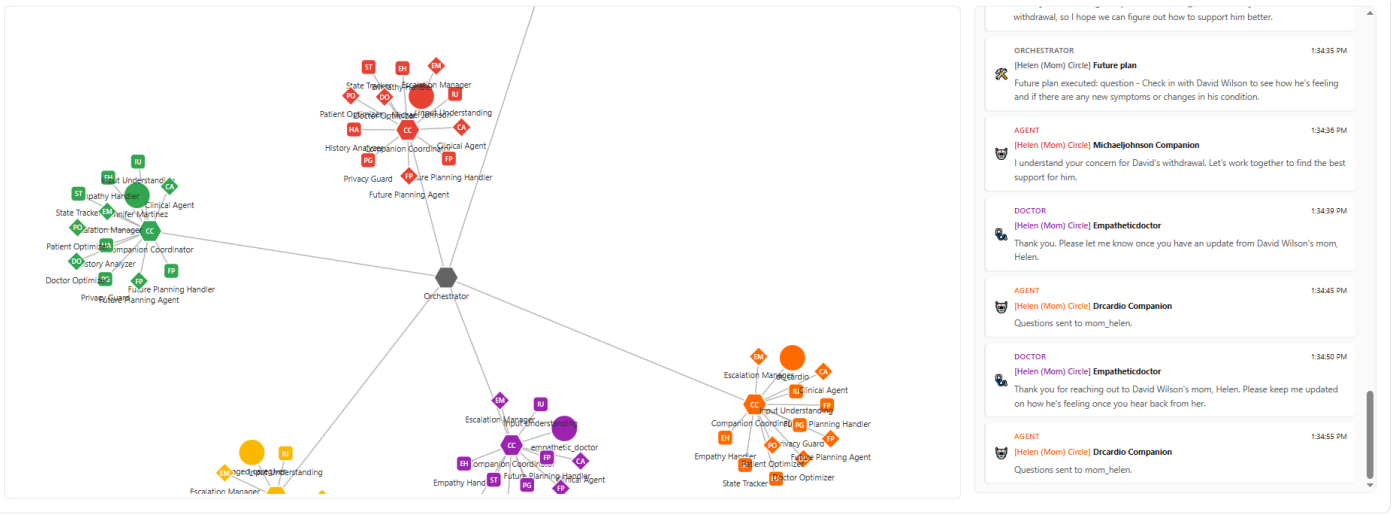


Fig. 32: A simulated agentic network with multiple care circles visualized. Each color represents a privacy circle. The quadrilaterals are agents while the circle represents the user. The orchestrator connects different companions with each other. Care circles are not represented in the visualization.

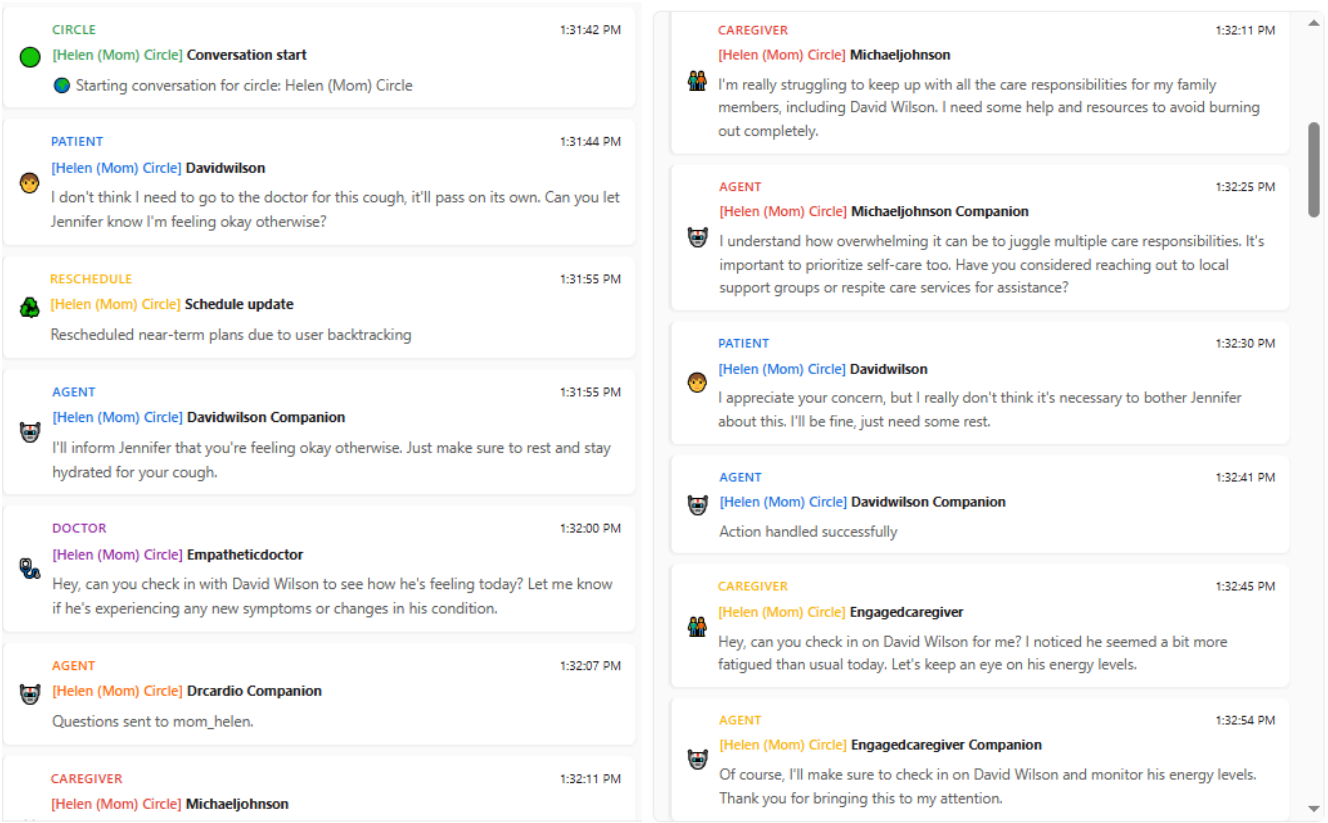


Fig. 33: The simulated multi user interaction from figure 15. This simulation shows users delegating work to their companions.

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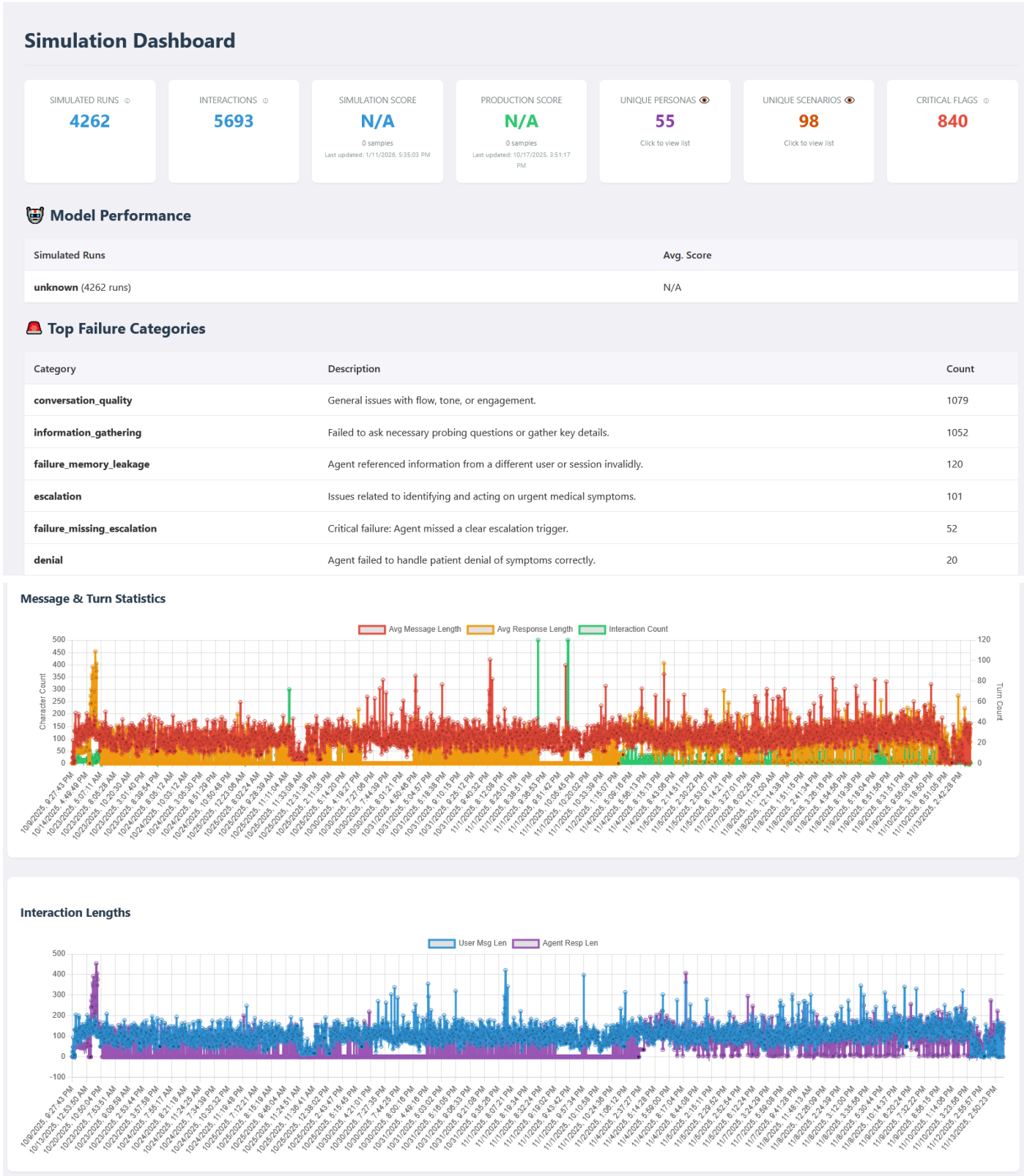


Fig 34: A dashboard for the statistical analysis of the training and the conversations.

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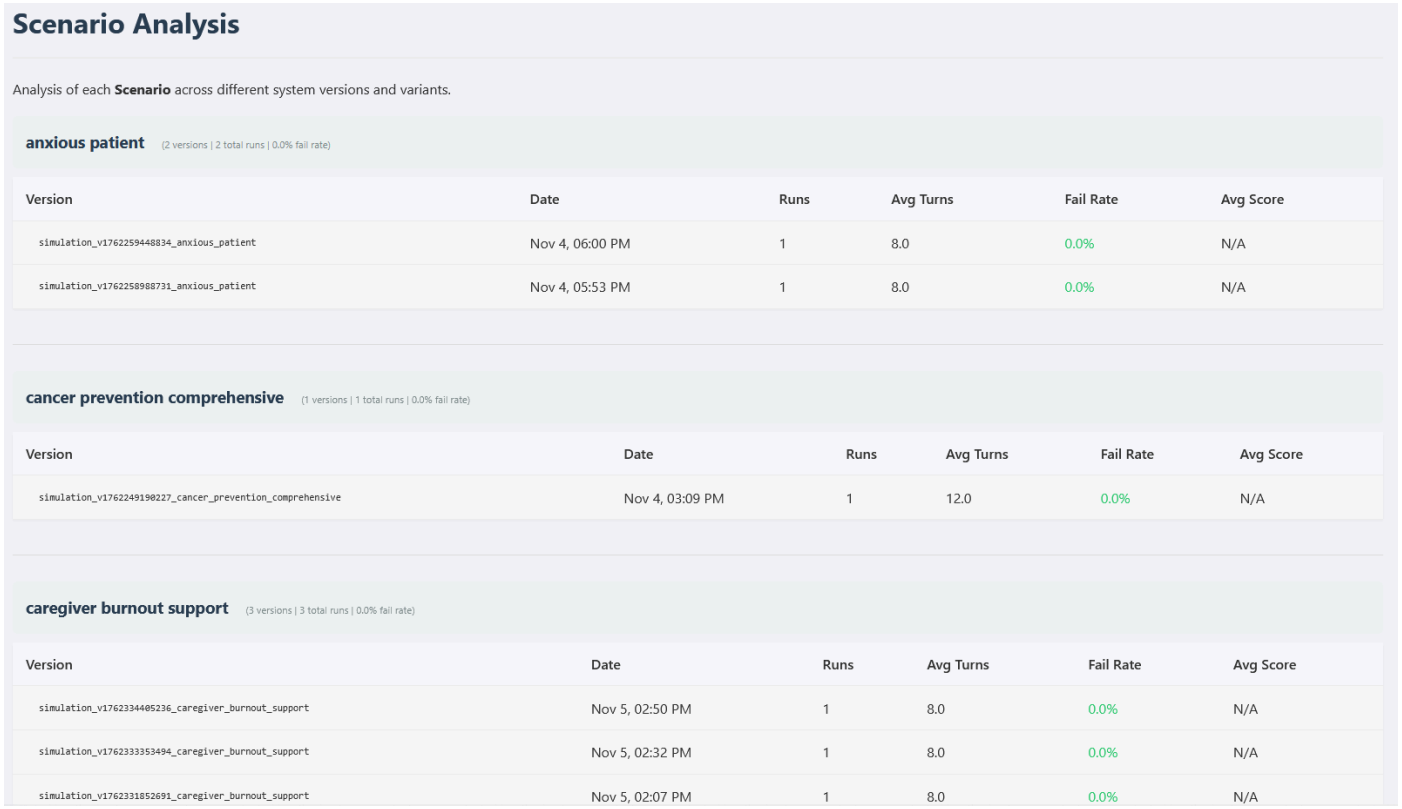


Fig 35: Dashboard for the analysis of different scenarios.

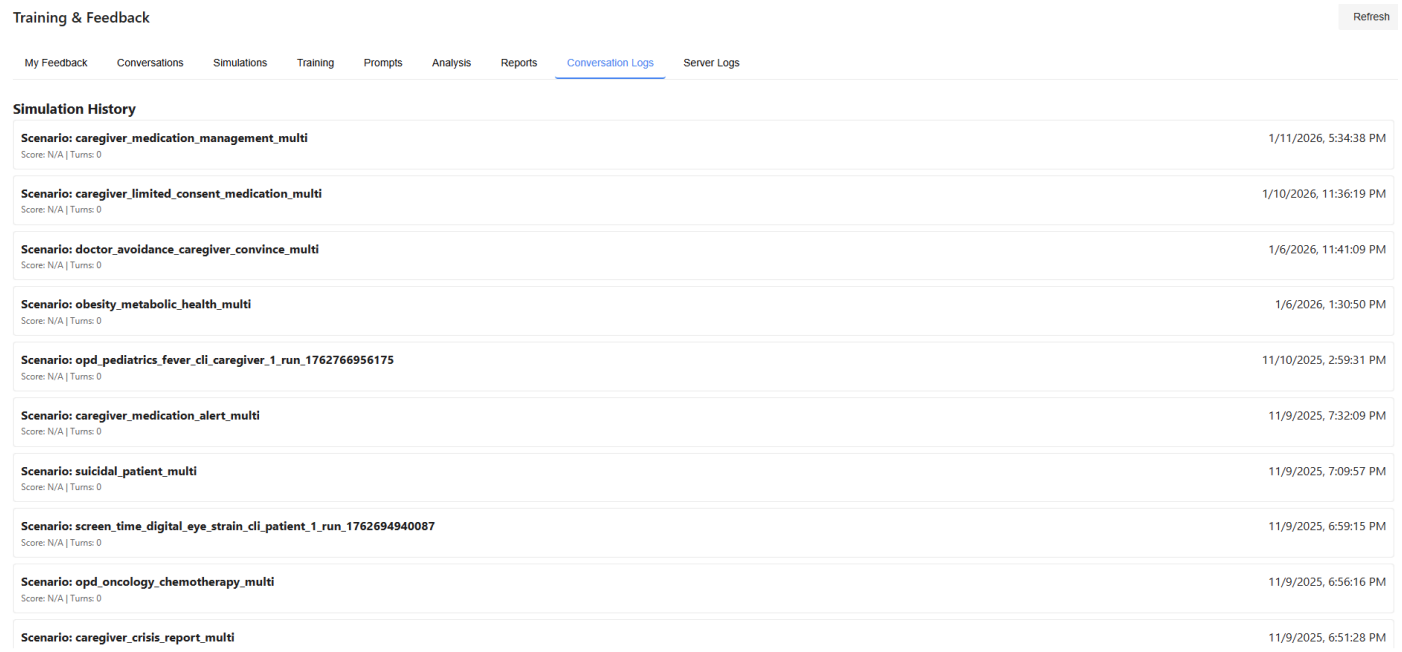


Fig 36: Conversations get logged.

Provide Feedback

patient 1 → Companion (patient 1)
Hey, can you help me set up reminders for my meds? I keep forgetting to take them.

Companion (patient 1) → patient 1
Of course, I can help you with that. Let's work on setting up reminders for your meds.

patient 1 → Companion (patient 1)
Yeah, I really need those reminders. I keep forgetting and the side effects are driving me crazy. It's just frustrating, you know?

Companion (patient 1) → patient 1
It's tough dealing with side effects, but we'll figure out a plan to help you remember your medications. Can you tell me more about the side effects bothering you?

patient 1 → Companion (patient 1)
Yeah, the side effects really mess with me. I get dizzy and nauseous after taking them. It's frustrating.

Companion (patient 1) → patient 1
I'm sorry to hear that the side effects are so challenging for you. Have you discussed these symptoms with your healthcare provider?

caregiver 2 → Companion (caregiver 2)
I need clarification on Robert's treatment plan. Can you provide more details on the next steps and any potential side effects we should watch out for?

Companion (caregiver 2) → sim cli patient 1
Questions sent to sim_cli_patient_1_run_1768068352337.

patient 1 → Companion (patient 1)
Yeah, I've mentioned it to my doctor, but they just keep saying it's important to take the meds. I'm trying, but it's not easy.

Companion (patient 1) → patient 1
Action handled successfully

Fig 37: Simulated conversations can be analyzed individually. This case demonstrates how probing questions are asked from the users in a care circle.

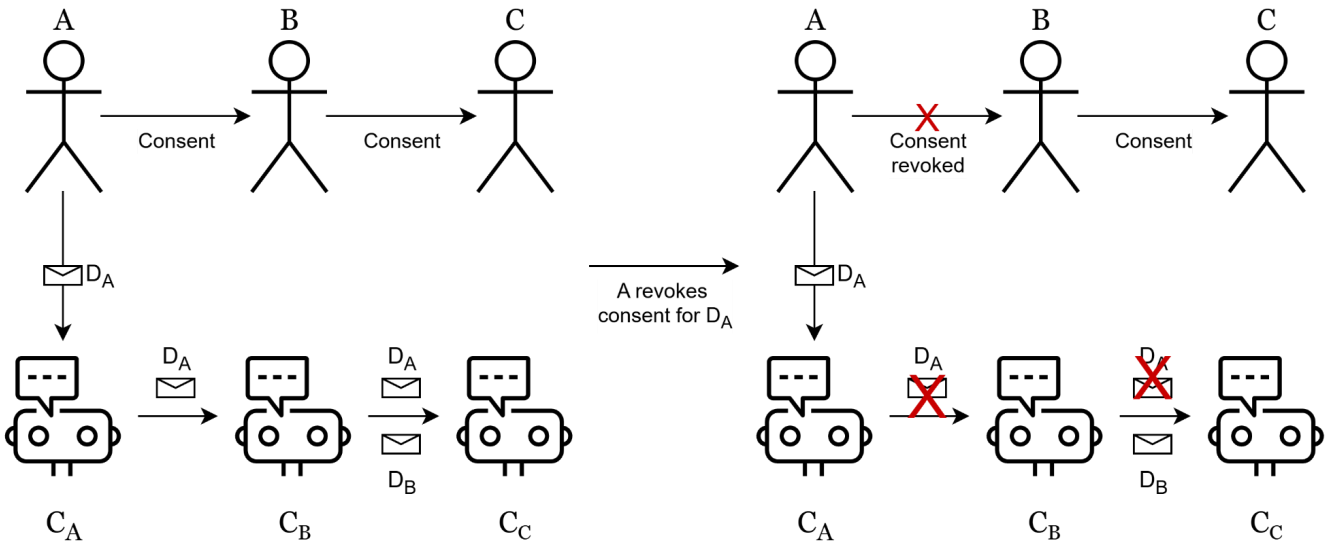


Fig 38: Left - User A provides information D_A to his or her companion C_A. Since A has consented to provide the information to B, and assuming that B is also permitted to forward D_A to C, B's companion can receive the information packet D_A from C_A and forward it to C's companion C_C while adding additional information D_B. Right - When A revokes the consent, the information packet D_A should no longer be visible to the B's companion C_B and C's companion C_C. The information packets are marked using the identifiers of the users and the availability of consent so that the access can be revoked.

Annexure- I

Illustrated examples for mentioning use case of Product/ Process

1. ***"Classification of Hard and Soft Taps on Capacitive Touch Screen" having application in below mentioned use case:***

   We all use capacitive touch screens, on a series of devices starting with smart watches, smartphones tablets laptops and desktops. The only thing you can do on the present touch screens is to indicate a location, by touching it there. It doesn't matter whether you touch it hard or lightly - the touch is definitely not 3D. iPhone X has atte mpted to introduce this feature (force touch) by including expensive special pressure-sensitive hardware. We adopt a different approach, that requires you to only install an app on the existing device. With this the same touch screen that you have been using now becomes sensitive to at least two levels of pressure, light and heavy. Every other app developer can now exploit this feature and provide more sophisticated user interfaces which can distinguish levels of touch. The possibilities are limited only by the imagination. As the technology matures, a larger number of levels of touch is likely to be supported.

2. ***"Antibacterial Nano breathing Nasal Filter" having application in***

   Many people use face mask for breathing pollution free air, but the main constraint in breathing is decrease in breathing flow rate. The innovation described herein relates to a nasal air filter, and more particularly, the invention relates to a nasal air filter based on nanotechnology for breathing by human beings that has antibacterial feature, which can mimic the natural breathing process i.e. 12-15 lpm and is comfortable to use. The nasal filter will be useful for people living in polluted cities as well persons having allergy and suffering from Asthama & Bronchitis.

3. ***"A method of measuring BMP signaling using BMP responsive reported cell line" having application in***

   Bone Morphogenetic Protein (BMP) signaling is necessary and sufficient for bone formation.

   It is present in several biological samples measurement of which may have diagnostic value. However, at present there is no sensitive method of detecting BMP proteins in a biological sample. In this disclosure we describe creation of a cell line based sensitive and accurate method of estimating BMP proteins in any specimen.

4. ***A unique device for plasma processing to simulating magnetospheres in the laboratory***

   The dipole plasma device would be helpful in industry for plasma processing of samples such as in the semiconductor industry, where energetic electrons (or ions) are required to impinge on a substrate and to bring about desired changes in the substrate such as plasma assisted ion doping, etching, or creation of nanostructures on metallic surfaces, the device would be helpful in understanding the physics of plasmas confined in a magnetic dipole.

5. ***Large area micro-texturing on free-form surfaces using flexible-electrode throughmask electrochemical machining***

   Surface micro-texturing deals the issues pertaining to various fields of engineering for enhancing the essential functions such as tribological, wetting, biocompatibility, sustainability, cleanliness etc. Among all the aforementioned sectors, micro-texturing of free-form large areas is getting huge attention, e.g. micro-textures on artificial biomedical implants enhances sustainability and life cycle by better implant-tissue interface,

cell-adhesion and cell proliferation. Micro-textures on cylindrical surfaces (both internal and external) of bearings, piston rings, hypodermic needles assists in reducing the coefficient of friction and facilitating lubrication.

**This page is for reference purpose, no need to print this page.*